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INFORMATION PROCESSING SYSTEM, ENDOSCOPE SYSTEM, IMAGE PROCESSING METHOD AND INFORMATION STORAGE MEDIUM

Abstract

Defocus simulation processing is performed for a region on an optical axis of a first imaging system and a region other than on the optical axis in a training image, based on a transfer function or a point spread function on the optical axis. One or more processors use a trained model to generate an output image in which a blur of a processing target image which is an image captured by the first imaging system is corrected, and estimates an object distance of the processing target image. The one or more processors acquire a filter characteristic associated with the estimated object distance from a correction table, and performs blur adjustment processing for the output image using the acquired filter characteristic.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS

[0001] This application is based upon and claims the benefit of priority to Japanese Patent Application No. 2024-035184 filed on Mar. 7, 2024, the entire contents of which are incorporated herein by reference.

BACKGROUND

[0002] In endoscopic observation, for example, magnifying observation closer to a subject is desired. Optically, however, a depth of field is narrower with a higher resolution with finer pixels. Thus, technologies for extending the depth of field using image processing techniques have been sought. WO 2018/037521 discloses a technology that corrects optical degradation of an imaging system by deep learning.

[0003] WO 2018/037521 uses, as a training image, a reference image captured in advance with optical degradation information. However, there are myriads of optical degradation information to learn depending on object distance and image height, and therefore an enormous number of training images are required. Consequently, the network scale necessary for processing increases, leading to reduction in processing ability and increase in implementation costs.

SUMMARY

[0004] In accordance with one of some aspect, there is provided an information processing system comprising: [0005] a memory configured to store a trained model trained by machine learning with a data set including a training image, a true image, and an object distance label, and a correction table in which an object distance is associated with a filter characteristic for blur correction, the object distance being a distance between an imaging system and a subject; and [0006] one or more processors, wherein [0007] the training image is generated by performing defocus simulation processing that simulates, for a predetermined subject image in focus captured by a given imaging system, an effect of a blur caused by defocus of a first imaging system, based on a transfer function or a point spread function of the first imaging system at a predetermined object distance, [0008] the defocus simulation processing is performed for a region on an optical axis of the first imaging system and a region other than on the optical axis in the training image, based on the transfer function or the point spread function on the optical axis, [0009] the true image is an image generated by performing best focus simulation processing that simulates, for the predetermined subject image, a state in which the first imaging system is focused, based on the transfer function or the point spread function at the object distance at which the first imaging system is focused, or the predetermined subject image itself, [0010] the trained model is trained by machine learning so that the training image is the true image, and trained by machine learning by applying, as the object distance label, the object distance of the transfer function or the point spread function of the first imaging system used in the defocus simulation processing, and [0011] the one or more processors [0012] use the trained model to generate an output image in which a blur of a processing target

image is corrected, the processing target image being an image captured by the first imaging system, [0013] estimate the object distance of the processing target image, and [0014] acquire, from the correction table, the filter characteristic associated with the estimated object distance, and perform blur adjustment processing for the output image using the acquired filter characteristic.
[0015] In accordance with one of some aspect, there is provided an endoscope system comprising: [0016] the information processing system according to any one of claims **1** to **9**; and [0017] an endoscopic scope configured to capture the processing target image.
[0018] In accordance with one of some aspect, there is provided an image processing method using a trained model trained by machine learning with a data set including a training image, a true image, and an object distance label, and a correction table in which an object distance is associated with a filter characteristic for blur correction, the object distance being a distance between an imaging system and a subject, wherein [0019] the training image is generated by performing defocus simulation processing that simulates, for a predetermined subject image in focus captured by a given imaging system, an effect of a blur caused by defocus of a first imaging system, based on a transfer function or a point spread function of the first imaging system at a predetermined object distance, [0020] the defocus simulation processing is performed for a region on an optical axis of the first imaging system and a region other than on the optical axis in the training image, based on the transfer function or the point spread function on the optical axis, [0021] the true image is an image generated by performing best focus simulation processing that simulates, for the predetermined subject image, a state in which the first imaging system is focused, based on the transfer function or the point spread function at the object distance at which the first imaging system is focused, or the predetermined subject image itself, [0022] the trained model is trained by machine learning so that the training image is the true image, and trained by machine learning by applying, as the object distance label, the object distance of the transfer function or the point spread function of the first imaging system used in the defocus simulation processing, [0023] the image processing method comprising: [0024] using the trained model to generate an output image in which a blur of a processing target image is corrected, the processing target image being an image captured by the first imaging system; [0025] estimating the object distance of the processing target image; and [0026] acquiring, from the correction table, the filter characteristic associated with the estimated object distance, and performing blur adjustment processing for the output image using the acquired filter characteristic.
[0027] In accordance with one of some aspect, there is provided a non-transitory information storage medium that stores a program for causing a computer to execute the image processing method according to claim **11**.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0028] FIG. **1** is a block diagram illustrating a configuration example of an information processing system.
[0029] FIG. **2** is a block diagram illustrating the configuration example of the information processing system in more detail.
[0030] FIG. **3** is a flowchart illustrating a processing example of the information processing system.
[0031] FIG. **4** is a block diagram illustrating a configuration example of a training device.
[0032] FIG. **5** is a diagram illustrating a training model.
[0033] FIG. **6** is a diagram illustrating a neural network.
[0034] FIG. **7** is a flowchart illustrating trained model creation processing.
[0035] FIG. **8** is a diagram illustrating machine learning according to an embodiment of the present

disclosure.
[0036] FIG. **9** is a diagram illustrating a relationship between a depth of field and a target depth of field.
[0037] FIG. **10** is a diagram illustrating image data generation processing according to an embodiment.
[0038] FIG. **11** is a diagram illustrating the image data generation processing according to another embodiment.
[0039] FIG. **12** is a diagram illustrating a transfer function or a point spread function.
[0040] FIG. **13** is a diagram illustrating defocus simulation processing according to an embodiment of the present disclosure.
[0041] FIG. **14** is a block diagram illustrating an endoscope system according to an embodiment.
[0042] FIG. **15** is a block diagram illustrating the endoscope system according to another embodiment.
[0043] FIG. **16** is a graph illustrating a relationship between an object distance and MTF in the defocus simulation processing.
[0044] FIG. **17** is another graph illustrating a relationship between an object distance and MTF in the defocus simulation processing.
[0045] FIG. **18** is a diagram illustrating a specific computation technique in the defocus simulation processing.
[0046] FIG. **19** is another diagram illustrating a specific computation technique in the defocus simulation processing.
[0047] FIG. **20** is a diagram illustrating a specific computation technique in best focus simulation processing.
[0048] FIG. **21** is another diagram illustrating a specific computation technique in the best focus simulation processing.
[0049] FIG. **22** is a diagram illustrating a lens configuration of a first imaging system according to an embodiment.
[0050] FIG. **23** is a block diagram illustrating another configuration example of the information processing system.
[0051] FIG. **24** is a block diagram illustrating another configuration example of the information processing system in more detail.
[0052] FIG. **25** is a flowchart illustrating another processing example of the information processing system.
[0053] FIG. **26** is a graph illustrating a relationship between an object distance and MTF in the defocus simulation processing.
[0054] FIG. **27** is another graph illustrating a relationship between an object distance and MTF in the defocus simulation processing.
[0055] FIG. **28** is a diagram illustrating a specific computation technique in the defocus simulation processing.
[0056] FIG. **29** is another diagram illustrating a specific computation technique in the defocus simulation processing.
[0057] FIG. **30** is a diagram illustrating a specific computation technique in best focus simulation processing.
[0058] FIG. **31** is another diagram illustrating a specific computation technique in the best focus simulation processing.
[0059] FIG. **32** is a diagram illustrating a lens configuration of a first imaging system according to an embodiment.
[0060] FIG. **33** is another diagram illustrating a lens configuration of the first imaging system according to another embodiment.
[0061] FIG. **34** is a diagram illustrating an amount of distortion.

[0062] FIG. **35** is a diagram illustrating a lens configuration including a phase modulation element.
[0063] FIG. **36** is a graph illustrating change in MTF due to inclusion of the phase modulation element according to an embodiment.
[0064] FIG. **37** is a diagram illustrating the image data generation processing according to another embodiment.
[0065] FIG. **38** is a diagram illustrating the defocus simulation processing according to another embodiment.
[0066] FIG. **39** is a diagram illustrating the image data generation processing according to another embodiment.
[0067] FIG. **40** is a diagram illustrating the defocus simulation processing according to another embodiment.
[0068] FIG. **41** is a diagram illustrating the best focus simulation processing according to another embodiment.
[0069] FIG. **42** is a diagram illustrating another configuration example of the information processing system.
[0070] FIG. **43** is a diagram illustrating the image data generation processing according to another embodiment.
[0071] FIG. **44** is a diagram illustrating the defocus simulation processing according to another embodiment.
[0072] FIG. **45** is a diagram illustrating a relationship between mosaicing processing and demosaicing processing.
[0073] FIG. **46** is a diagram illustrating the best focus simulation processing according to another embodiment.
[0074] FIG. **47** is a diagram illustrating another configuration example of the information processing system.
[0075] FIG. **48** is a flowchart illustrating another processing example of the information processing system.
[0076] FIG. **49** is a flowchart illustrating first trained model creation processing.
[0077] FIG. **50** is a flowchart illustrating second trained model creation processing.
[0078] FIG. **51** is a diagram illustrating the image data generation processing according to another embodiment.
[0079] FIG. **52** is a diagram illustrating the defocus simulation processing according to another embodiment.
[0080] FIG. **53** is a diagram illustrating the best focus simulation processing according to another embodiment.
[0081] FIG. **54** is a diagram illustrating evaluation of a frequency characteristic.
[0082] FIG. **55** is a diagram illustrating the relationship between a first frequency characteristic and a second frequency characteristic, and the like.
[0083] FIG. **56** is a diagram illustrating a modification of correction processing.
[0084] FIG. **57** is a diagram illustrating adjustment of a blur for an output image in more detail.
[0085] FIG. **58** is another diagram illustrating adjustment of a blur for an output image in more detail.
[0086] FIG. **59** is a diagram illustrating another configuration example of the trained model.

DESCRIPTION OF EXEMPLARY EMBODIMENTS

[0087] The following disclosure provides many different embodiments or examples for implementing different features of the provided subject matter. These are, of course, merely examples and are not intended to be limiting. In addition, the disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed. Further, when a first element is described as being “connected” or "coupled" to a second element, such description includes embodiments in which the first and second elements are directly connected or coupled to each other, and also includes embodiments in which the first and second elements are indirectly connected or coupled to each other with one or more other intervening elements in between.

[0088] FIG. **1** is a block diagram illustrating a configuration example of an information processing system **100** according to an embodiment of the present disclosure. The present information processing system **100** includes a memory section **110** and a processing section **130**. The memory section **110** stores a trained model **120** trained by machine learning. The trained model **120** is a program module that outputs a corrected image in which a blur caused by defocus in a processing target image is corrected. The trained model **120** is generated or updated by performing machine learning described later. The processing target image is, for example, image data captured by a first imaging system **101** as illustrated in FIG. **1**, but not limited thereto. The details will be described later. In the present embodiment, image data that can be processed as digital data may be simply referred to as image. A training image group **32**G is a set of training images **32** including a first training image **32-1**, a second training image **32-2**, . . . , and an Nth training image **32-**N. The training image group **32**G will be detailed later in conjunction with a true image **36**. In other words, the processing section **130** in the present embodiment uses the trained model **120** to correct a blur caused by defocus of the first imaging system **101** in the processing target image which is an image captured by the first imaging system **101**. The memory section **110** and the processing section **130** are also referred to as a memory device and a processing device, respectively.

[0089] When the information processing system **100** is incorporated into an endoscope system **300** described later with reference to FIG. **14** and the like, the training image **32** in the present embodiment is, for example, but not limited to, an image of biological tissue or an image of a subject that imitates biological tissue. For example, in a learning phase described later, an image of a chart that conforms to a predetermined standard, such as a spatial frequency response (SFR) chart, may be employed as the training image **32**. An image of a chart with a predetermined modification to such a chart image may also be employed as the training image **32**. The predetermined modification is, for example, a modification that makes the contrast of the chart common to the contrast of biological tissue.

[0090] Machine learning in the present embodiment is, for example, supervised learning. Training data in supervised learning is a data set in which input data is associated with a ground truth label. Specifically, the trained model **120** in the present embodiment is generated by supervised learning based on a data set in which input data including the training images **32** simulating the effects of various kinds of blur is associated with a ground truth label including the true image **36** in focus.

[0091] The processing section **130** in the present embodiment is configured with hardware described below. The hardware can include at least one of a circuit that processes digital signals and a circuit that processes analog signals. For example, the hardware can be configured with one or more circuit devices or one or more circuit elements mounted on a circuit board. One or more circuit devices are, for example, ICs. One or more circuit elements are, for example, capacitors.

[0092] The processing section **130** may be implemented, for example, by a processor described below. The processing section **130** in the present embodiment includes a memory that stores information and a processor that operates based on the information stored in the memory. The memory is, for example, the memory section **110**. The information is, for example, a program and various data. The processor includes hardware. Various processors such as a central processing unit (CPU), a graphics processing unit (GPU), and a digital signal processor (DSP) can be used as the processor. The memory may be a semiconductor memory such as a static random access memory (SRAM) or a dynamic random access memory (DRAM), a register, a magnetic storage device such as a hard disc device, or an optical storage device such as an optical disc device. For example, the memory stores computer-readable instructions, and the instructions are executed by the processor to implement the function of each unit of the processing section **130** as processing. As used herein

the instructions may be instructions of an instruction set that constitutes a program or may be instructions to instruct the hardware circuit of the processor to operate.
[0093] Further, the trained model **120** in the present embodiment may be used by the information processing system **100** depicted in a configuration example in FIG. **2**. In other words, the trained model **120** in the present embodiment is used by the information processing system **100** including the memory section **110** that stores the trained model **120**, an input section **140**, the processing section **130**, and an output section **150**, and is trained by machine learning using a data set including the training image group **32**G and the true image **36**.
[0094] The input section **140** is an interface that receives the processing target image from outside. Specifically, for example, as illustrated in FIG. **1** and FIG. **2**, the input section **140** is an image data interface that receives image data as the processing target image from the first imaging system **101**. For example, the input section **140** inputs the received processing target image as input data to the trained model **120** and the processing section **130** performs processing described later, whereby the function as the input section **140** is served. In other words, in the trained model **120** in the present embodiment, the input section **140** inputs the processing target image, which is an image captured by the first imaging system **101**, to the trained model **120**.
[0095] The output section **150** is an interface that transmits the corrected image described above to the outside. For example, output data from the trained model **120** is used as the corrected image transmitted by the output section **150**, whereby the function as the output section **150** is served. The corrected image is transmitted to, for example, a predetermined display device connected to the information processing system **100**. For example, the output section **150** is an interface that can be connected to the predetermined display device so that the corrected image appears on the display device, whereby the function as the output section **150** is served. The corrected image may be output to, for example, a storage device which is an external device.
[0096] FIG. **3** is a flowchart illustrating a technique performed by the information processing system **100** according to the present embodiment. The processing section **130** reads the processing target image (step S**10**) and reads the trained model (step S**20**), and thereafter performs correction processing (step S**30**). Specifically, for example, the processing section **130** performs processing of inputting the processing target image received via the input section **140** to the trained model **120** read from the memory section **110**. If it is determined that the processing target image, which is input data, is common to the training image **32**, the trained model **120** estimates that data to be output is the true image **36**. Thus, upon input of the processing target image, the trained model **120** outputs the true image **36**. When the processing target image and the true image **36** are compared, there is a relationship such that the true image **36** is an image in which a blur caused by defocus of the first imaging system **101** in the processing target image is corrected. In other words, the processing section **130** performs the correction processing of correcting a blur caused by defocus of the first imaging system **101** in the processing target image, using the trained model **120** (step S**30**).
[0097] The processing section **130** thereafter outputs the corrected image (step S**40**). Specifically, the output section **150** functions as described above whereby the corrected image is output to a desired destination. In other words, the output section **150** outputs the corrected image produced by the correction processing.
[0098] The machine learning of the trained model **120** will now be described. The machine learning is performed, for example, by a training device **10**. FIG. **4** is a block diagram illustrating a configuration example of the training device **10**. The training device **10** includes, for example, a communication section **12**, a training device processing section **16**, and a training device memory section **18**.
[0099] The communication section **12** is a communication interface that can communicate with the information processing system **100** via a predetermined communication scheme. The predetermined communication scheme is, for example, a communication scheme in conformity with a wireless communication standard such as Wi-Fi (registered trademark), but not limited

thereto, and may be a communication scheme in conformity with a wired communication standard such as USB. With this configuration, the training device **10** can transmit the trained model **120** trained by machine learning according to a technique described later to the information processing system **100**, and the information processing system **100** can update the trained model **120**. Although FIG. **4** illustrates an example in which the training device **10** is separate from the information processing system **100**, it is not intended to preclude a configuration example in which the information processing system **100** includes a training server corresponding to the training device **10**.

[0100] The training device processing section **16** performs input/output control of data to/from each functional unit such as the communication section **12** and the training device memory section **18**. The training device processing section **16** can be implemented by a processor similar to the processing section **130** in FIG. **1**. The training device processing section **16** controls operation such as data output to the information processing system **100** by executing various computation processing based on a predetermined program read from the training device memory section **18** and an operation input signal and the like from an operation unit not illustrated in FIG. **4**. The predetermined program here includes a machine learning program. In other words, the training device processing section **16** serves the function of machine learning by reading the machine learning program, necessary data, and the like from the training device memory section **18** and executing the machine learning program.

[0101] The training device memory section **18** stores a training model **20**, a predetermined subject image **30**, and optical system information **40**, in addition to the not-illustrated machine learning program. The training device memory section **18** can be implemented by, for example, a semiconductor memory similar to the memory section **110** described above. The training device memory section **18** may further include another information. Another information is, for example, image sensor information **50** described later.

[0102] The predetermined subject image **30** is an image of a subject associated with the processing target image. The training image **32** and the true image **36**, which will be described later, are created based on the predetermined subject image **30**. In other words, the training device memory section **18** stores predetermined subject images **30** as many as the kinds of subjects from which processing target images can be produced. To give a more specific example, in a case where the information processing system **100** is used for an endoscope system **300** described later, a captured image of a lumen or the like captured by an endoscopic scope **310** described later is the predetermined subject image **30**. In the following description, an imaging system that captures the predetermined subject image **30** is referred to as a given imaging system **104**, unless otherwise specified. A case where the predetermined subject image **30** is captured with a specified imaging system will be described later.

[0103] The training model **20** is a model subjected to machine learning by the training device processing section **16**. The model here is information that derives a correspondence between estimation target data and estimation result data. More specifically, the model is information that derives an output image **34** which is the estimation result data from the training image **32** which is the estimation target data. In the training model **20** in the present embodiment, a neural network NN is included in at least a part of the model. The detail of the neural network NN will be described later with reference to FIG. **6**. In the case where the information processing system **100** and the training device **10** are integrated as described above, the trained model **120** may be subjected to machine learning.

[0104] The “output image” in the present embodiment refers to an image output from the training model **20** in the learning phase, or an image output from a training model **1020** described later, but may also include an image output from a trained model **1120** in an inference phase in an example described later. Hereinafter, for convenience of explanation, an image output from the training model **20** or the training model **1020** is referred to as output image **34**, and an image output from the trained model **1120** described later is referred to as output image **134**.

[0105] For example, when the first training image **32-1** is input to the training model **20**, the training model **20** outputs a first output image **34-1**. Similarly, when the Nth training image **32-**N is input to the training model **20**, the training model **20** outputs an Nth output image **34-**N. In other words, as illustrated in FIG. **5**, in the training device **10** according to the present embodiment, N images including the first training image **32-1** to the Nth training image **32-**N are input as the training image group **32**G to the training model **20**.

[0106] FIG. **6** is a diagram illustrating the neural network NN. The neural network NN includes an input layer that receives data, an intermediate layer that performs computation based on an output from the input layer, and an output layer that outputs data based on an output from the intermediate layer. FIG. **6** illustrates a network including two intermediate layers by way of example, but the network may include one intermediate layer or three or more intermediate layers. The number of nodes included in each layer is not limited to the number in the example in FIG. **6**, and a variety of modifications may be implemented. As illustrated in FIG. **6**, nodes included in a given layer are connected to nodes in an adjacent layer. A weighting factor is set for each connection. Each node multiplies an output from a node in the preceding layer by the weighting factor and obtains a sum of multiplication results. In addition, each node adds a bias to the sum and applies an activation function to the addition result to obtain an output of the node. This processing is successively executed from the input layer toward the output layer, resulting in an output of the neural network NN. Various functions such as sigmoid and ReLU functions are known as the activation function and can be widely applied in the present embodiment.

[0107] Models with various configurations are known for the neural network NN and can be widely applied in the present embodiment. For example, the neural network NN may be a convolutional neural network (CNN), a recurrent neural network (RNN), or other models.

[0108] FIG. **7** is a flowchart illustrating a processing example of trained model creation processing (step S**100**). The trained model creation processing (step S**100**) is processing of creating or updating the trained model **120** by machine learning. The training device processing section **16** reads the predetermined subject image (step S**110**) and thereafter performs image data generation processing (step S**120**). For example, the training device processing section **16** reads the predetermined subject image **30** from the training device memory section **18** and performs predetermined processing for generating the training image **32** and the true image **36** using the predetermined subject image **30**. The predetermined processing is, for example, defocus simulation processing (step S**200**) and best focus simulation processing (step S**300**), which will be detailed later.

[0109] The training device processing section **16** thereafter performs correction learning processing (step S**130**). For example, the training device processing section **16** performs processing of reading the training model **20** from the training device memory section **18**, processing of inputting the training image **32** generated in the image data generation processing (step S**120**) to the training model **20**, and machine learning processing based on the output image **34** output from the training model **20** and the true image **36**.

[0110] The machine learning processing based on the output image **34** and the true image **36** is processing of changing a network parameter of the neural network NN so that the first output image **34-1** to Nth output image **34-**N are the true image **36**, for example, as illustrated in FIG. **8**. The processing of changing a network parameter of the neural network NN is specifically, for example, processing of updating an appropriate weighting factor in the neural network NN. The weighting factor here includes a bias. In the updating of the weighting factor, for example, back propagation can be used, which updates the weighting factor from the output layer toward the input layer. In other words, the training device **10** inputs input data of training data to a model and obtains an output by performing forward computation in accordance with a model configuration using the weighting factor at that moment. An error function is calculated based on the output and

the ground truth label, and the weighting factor is updated so that the error function is reduced.
[0111] More specifically, for example, the training device processing section **16** inputs the first training image **32-1** as input data to the neural network NN included in the training model **20**, and outputs the first output image **34-1** as output data by performing computation in the forward direction using the weighting factor at that moment. The training device processing section **16** computes the error function, based on the first output image **34-1** and the true image **36** which is the ground truth label. The training device processing section **16** then performs processing of updating the weighting factor so that the error function is reduced. Further, the training device processing section **16** repeatedly performs similar processing for the second output image **34-2** to the Nth output image **34**-N. In this way, the training model **20** is trained by machine learning so that one true image **36** can be output for different kinds of training images **32**. With this configuration, the training model **20** trained by machine learning is output as the trained model **120** to the information processing system **100**, whereby the trained model **120** stored in the memory section **110** is updated. Although FIG. **4** illustrates the training device **10** and the information processing system **100** communicatively connected via the communication section **12**, the training device **10** and the information processing system **100** are not necessarily communicatively connected. In this case, for example, a user performs processing on the training device **10** to temporarily store the training model **20** as the trained model **120** into an information storage medium, and brings the information storage medium to where the information processing system **100** is present, and performs processing on the information processing system **100** to update the trained model **120** based on the information storage medium, thereby implementing updating of the trained model **120**.
[0112] FIG. **9** is a diagram illustrating a relationship between a focal depth and a depth of field for the first imaging system **101** in the present embodiment, where the optical axis is the horizontal axis. FIG. **9** is an illustration for convenience and does not depict a specific lens configuration of the first imaging system **101**. For example, in FIG. **9**, the range indicated by DP**1** is a depth of field corresponding to the focal depth in optical design of the first imaging system **101**. Therefore, for example, when the distance between a subject and the first imaging system **101** is a first object distance indicated by D**1**, the subject is located outside the range of the depth of field and therefore a processing target image captured with the first imaging system **101** includes the effect of a blur caused by defocus. For example, when the distance between the subject and the first imaging system **101** is a second object distance indicated by D**2**, the subject is located inside the depth of field, resulting in a processing target image in focus. For example, when the distance between the subject and the first imaging system **101** is an object distance indicated by D**3**, that is, a position indicated by P**1** on the optical axis in the depth of field is a position that satisfies a best focus condition. In FIG. **9**, the first object distance indicated by D**1** and the second object distance indicated by D**2** are located on the near-point side relative to the position indicated by P**1**, but may be on the far-point side rather than the near-point side. In the following description and illustration, techniques according to the present embodiment will be described using the object distance on the near-point side and the like as an example. However, it is not intended to preclude the techniques according to the present embodiment from being applied using the object distance on the far-point side and the like.
[0113] For example, in a system including the first imaging system **101**, there is a need for extending the depth of field because increasing a resolution with finer pixels narrows the depth of field. Further, for example, when the first imaging system **101** is used in the endoscopic scope **310** of the endoscope system **300** described later, the operation of adjusting the endoscopic scope **310** to a best focus position for a desired subject involves difficulty. Thus, there is a need for extending the depth of field.
[0114] Then, in the present embodiment, the information processing system **100** incorporates the trained model **120** trained by machine learning described above with reference to FIG. **8** and the

like, using a data set including an image that simulates the effect of a blur for the predetermined subject image **30** captured in advance as the training image **32**, and an image in focus as the true image **36**. In this manner, the processing in FIG. **3** is performed for the captured image with the effect of a blur caused by defocus as the processing target image, whereby the corrected image in focus is output from the information processing system **100**. As a result, the range of depth of field of the first imaging system **101** can be substantially extended.

[0115] More specifically, the depth of field can be substantially extended from the range indicated by DP**1** in FIG. **9** to the range indicated by DP**2**. As used herein "substantially extend" means that the depth of field is optically not extended but the depth of field is apparently extended to a range in which an image of a subject essentially located outside the range of depth of field is captured as if it was located within the range of depth of field, through image processing performed by the information processing system **100**. In other words, when the subject is located at a position away from the first imaging system **101** by the object distance indicated by D**1**, a processing target image having a blur is output from the first imaging system **101**. However, since the position is located within the range of substantial depth of field indicated by DP**2**, the processing target image is corrected to a corrected image in focus, and the corrected image is output from the information processing system **100**. In the following description, the substantial depth of field indicated by DP**2** in FIG. **9** that is extended using the trained model **120** in the present embodiment is referred to as target depth of field. The corrected image in focus here is not necessarily in focus in a strict sense. For example, even when the output corrected image is partially blurred, the user may determine that the function of the information processing system **100** suffices as long as, for example, treatment using the endoscopic scope **310** is practicable. In other words, the distance of the target depth of field in the present embodiment is a distance that is wider than the distance of depth of field optically defined but is variable according to the user's acceptance level. Therefore, DP**2** in FIG. **9** is depicted for the sake of convenience and is not intended to depict a constant length. This is applicable in the following description.

[0116] The trained model **120** in the present embodiment is trained by machine learning such that a blurred image obtained by capturing an image of a subject located in the range indicated by DP**10** in FIG. **9**, as a difference between the target depth of field indicated by DP**2** and the depth of field indicated by DP**1**, is corrected to an image in focus. In other words, the distance indicated by DP**10** is a distance necessary for machine learning.

[0117] The technique in the image data generation processing (step S**120**) for generating the training image **32** and the true image **36** necessary for the machine learning will now be described with reference to FIG. **10**. The technique in the image data generation processing is not limited to that of FIG. **10** and various modifications may be implemented as described later. The image data generation processing illustrated in FIG. **10** can also be called step S**120-1**.

[0118] It is assumed that the predetermined subject image **30** in the present embodiment is captured at an object distance at which an imaging system that captures the image is focused, in any of examples.

[0119] The training device processing section **16** generates the training image **32** by performing the defocus simulation processing (step S**200**) for the predetermined subject image **30** captured by the given imaging system **104**. In the following description, the defocus simulation processing for generating, for example, the first training image **32-1** can also be called step S**200-1**. Similarly, the defocus simulation processing for generating the Nth training image **32**-N can also be called step S**200**-N. This is applicable to steps S**202**, S**204**, S**206**, S**208**, S**210**, S**220**, and S**230** described later. For example, in generating the first training image **32-1** through the defocus simulation processing (step S**200-1**), the training device processing section **16** selects information on the first object distance from the read optical system information **40**. Similarly, in generating the second training image **32-2** in step S**200-2**, the training device processing section **16** selects information on the second object distance from the read optical system information **40**. In other words, in the present embodiment, this can be generalized to the following representation: the optical system information **40** corresponding to the Nth training image **32**-N is the Nth object distance, and in generating the Nth training image **32**-N, the training device processing section **16** selects information on the corresponding Nth object distance from the optical system information **40**. In the following description, the defocus simulation processing will be described taking the processing for generating the first training image **32-1** as an example, but the processing is similar to a case where the second training image **32-2** to the Nth training image **32**-N are generated.

[0120] Further, the training device processing section **16** generates the true image **36** by performing the best focus simulation processing (step S**300**) for the predetermined subject image **30**. For example, the training device processing section **16** selects information on an object distance at which the first imaging system **101** is focused, from the read optical system information **40**. The information on the object distance at which the first imaging system **101** is focused is a distance in design from the first imaging system **101** to the point indicated by P**1** in FIG. **9**, for example, as indicated by D**3**, which is an object distance corresponding to what is called a best focus condition.

[0121] The image data generation processing in the present embodiment may be as illustrated in FIG. **11**. The image data generation processing illustrated in FIG. **11** can also be called step S**120-2**. Processing similar to that of FIG. **10** will not be further elaborated.

[0122] Step S**120-2** in FIG. **11** differs from step S**120-1** in FIG. **10** in that the best focus simulation processing (step S**300**) is not performed, and that the true image **36** is the predetermined subject image **30** itself. This is because if the predetermined subject image **30** is an image captured at an object distance at which the given imaging system **104** is focused, the predetermined subject image **30** can be used as the true image **36**.

[0123] The defocus simulation processing (step S**200**) will be described with reference to FIG. **12** and FIG. **13**. The optical system information **40** to be read in performing the defocus simulation processing (step S**200**) includes information on a transfer function or a point spread function. The transfer function or the point spread function changes depending on an amount of defocus in the optical axis direction and an image height in a plane perpendicular to the optical axis. For example, at the first object distance, it is assumed that a region in a direction perpendicular to the optical axis and with the same size as the predetermined subject image **30** is divided into regions FC**11-1**, FC**12-1**, FC**13-1**, FC**21-1**, FC**22-1**, FC**23-1**, FC**31-1**, FC**32-1**, and FC**33-1**. In this case, the transfer function or the point spread function at the first object distance may exhibit a value different for each of the divided regions. Similarly, for example, at the Nth object distance, it is assumed that a region in a direction perpendicular to the optical axis and with the same size as the predetermined subject image **30** is divided into regions FC**11**-N, FC**12**-N, FC**13**-N, FC**21**-N, FC**22**-N, FC**23**-N, FC**31**-N, FC**32**-N, and FC**33**-N. In this case, the transfer function or the point spread function at the Nth object distance may exhibit a value different for each of the divided regions. Further, the transfer function or the point spread function of the region FC**11-1** and the transfer function or the point spread function of the region FC**11**-N may exhibit different values. This is applicable to cases between the regions FC**12-1** and FC**12**-N, . . . , between the regions FC**33-1** and FC**33**-N. In this way, if the training image group **32**G is a set of N training images **32**, as illustrated in FIG. **12**, information on transfer function or point spread function is enormous in performing machine learning.

[0124] In this respect, in the present embodiment, the transfer function or the point spread function on the optical axis is used in performing machine learning. In the present embodiment, it is assumed that the region FC**22-1** is a region where the optical axis of the first imaging system **101** passes. In other words, the transfer function or the point spread function in the region FC**22-1** is the transfer function or the point spread function on the optical axis of the first imaging system **101** at the first object distance. Similarly, the transfer function or the point spread function in the region FC**22**-N is the transfer function or the point spread function on the optical axis of the first imaging system **101** at the Nth object distance. The transfer function or the point spread function is divided into nine parts in FIG. **12**, but this is illustrated only by way of example. This is applicable to FIG. **13**. For example, the regions FC**22-1** to FC**22**-N in FIG. **12** are a set including a predetermined number of pixels in each of the vertical direction and the horizontal direction but may be one pixel. In other words, the transfer function or the point spread function on the optical axis in the present embodiment is the transfer function or the point spread function in at least one of the area of one pixel passing through the optical axis or the area of a predetermined number of pixels including the one pixel.

[0125] As illustrated in FIG. **13**, in the defocus simulation processing (step S**200**), processing of simulating the effect of a blur (step S**210**) is performed for the predetermined subject image **30**, based on the transfer function on the optical axis or the point spread function on the optical axis of the first imaging system **101**. Step S**210** will be detailed later. In other words, step S**210** is also performed for a region other than the region on the optical axis of the predetermined subject image **30**, based on the transfer function on the optical axis or the point spread function on the optical axis of the first imaging system **101**. For example, it is assumed that the predetermined subject image **30** is divided into nine regions, namely, regions AR**11**, AR**12**, AR**13**, AR**21**, AR**22**, AR**23**, AR**31**, AR**32**, and AR**33**, in the same manner as in FIG. **12**. For example, in a case where the first training image **32-1** is generated, the training device processing section **16** performs computation in step S**210-1** for the region AR**11** using the transfer function or the point spread function on the optical axis indicated in FC**22-1** in FIG. **12**. This computation is briefly referred to as AR**11***FC**22-1** in the following description and the illustration in FIG. **13**. This is applicable to computation in step S**210** and the like using other regions. Here "*" denotes convolution, for example, when PSF is used as the point spread function, which will be detailed later. Further, for example, when OTF is used as the transfer function, "*" means that a frequency characteristic of Fourier transform of the region AR**11** is multiplied by OTF of the region FC**22-1**.

[0126] In addition, the training device processing section **16** also performs step S**210-1** for the regions AR**12** to AR**33**, using the transfer function or the point spread function on the optical axis indicated in FC**22-1**. In other words, the training device processing section **16** performs AR**12***FC**22-1**, AR**13***FC**22-1**, AR**21***FC**22-1**, AR**22***FC**22-1**, AR**23***FC**22-1**, AR**31***FC**22-1**, AR**32***FC**22-1**, and AR**33***FC**22-1**, which are partially omitted in FIG. **13**. In this way, the training device processing section **16** divides the same region as the predetermined subject image **30** into a desired number of regions and performs step S**210** using the transfer function or the point spread function of one of the divided regions.

[0127] Similarly, it is assumed that the generated first training image **32-1** is divided into nine regions, namely, regions BR**11-1**, BR**12-1**, BR**13-1**, BR**21-1**, BR**22-1**, BR**23-1**, BR**31-1**, BR**32-1**, and BR**33-1**. The region BR**11-1** corresponds to the result of step S**210-1** performed for the region AR**11** described above. In other words, as illustrated in FIG. **13**, BR**11-1**=AR**11***FC**22-1**. Similarly, BR**12-1**=AR**12***FC**22-1**, BR**13-1**=AR**13***FC**22-1**, BR**21-1**=AR**21***FC**22-1**, BR**22-1**=AR**22***FC**22-1**, BR**23-1**=AR**23***FC**22-1**, BR**31-1**=AR**31***FC**22-1**, BR**32-1**=AR**32***FC**22-1**, and BR**33-1**=AR**33***FC**22-1**.

[0128] This technique is applicable to a case where the Nth training image **32**-N is generated. In other words, although not illustrated in the drawings, the training device processing section **16** performs BR**11**-N=AR**11***FC**22**-N, BR**12**-N=AR**12***FC**22**-N, . . . , BR**22**-N=AR**22***FC**22**-N, . . . , BR**32**-N=AR**32***FC**22**-N, and BR**33**-N=AR**33***FC**22**-N. Based on the foregoing, the defocus simulation processing (step S**200**) is performed for the region (BR**22**) on the optical axis of the first imaging system **101** and the regions other than on the optical axis (BR**11**, . . . , BR**21**, BR**23**, ... BR**33**) in each training image **32**, based on the transfer function or the point spread function on the optical axis (FC**22**).

[0129] The transfer function in the present embodiment may be called optical transfer function or OTF. OTF is an abbreviation of optical transfer function. The point spread function in the present embodiment may be called point spread function or PSF. PSF is an abbreviation of point spread function.

[0130] OTF is the result of Fourier transform of PSF. In other words, PSF is the result of inverse Fourier transform of OTF. Further, OTF is a complex function, and the absolute value of OTF is referred to as modulation transfer function, amplitude transfer function, or MTF. MTF is an abbreviation of modulation transfer function.

[0131] Based on the foregoing, the information processing system **100** according to the present embodiment includes the memory section **110** that stores the trained model **120** trained by machine learning with a data set including the training image group **32**G and the true image **36**, and the processing section **130** that uses the trained model **120** to correct a blur caused by defocus of the first imaging system **101** in the processing target image which is an image captured by the first imaging system **101**. The training image group **32**G includes a plurality of training images **32** generated by performing the defocus simulation processing (step S**200**) that simulates, for the predetermined subject image **30** in which the given imaging system **104** is focused on a predetermined subject of which image is captured by the given imaging system **104**, the effect of a blur caused by defocus of the first imaging system **101**, based on the transfer function or the point spread function of the first imaging system **101** at a plurality of object distances. The defocus simulation processing is performed for the region on the optical axis of the first imaging system **101** and the regions other than on the optical axis in each training image **32** of a plurality of training images **32**, based on the transfer function or the point spread function on the optical axis. The true image **36** is an image generated by performing the best focus simulation processing (step S**300**) that simulates, for the predetermined subject image **30**, a state in which the first imaging system **101** is focused, based on the transfer function or the point spread function at an object distance at which the first imaging system **101** is focused, or the predetermined subject image **30** itself. The trained model **120** is trained by machine learning so that each training image **32** is the true image **36**.

[0132] In this way, the information processing system **100** according to the present embodiment, which includes the memory section **110** that stores the trained model **120** and the processing section **130**, can output a corrected image in which the effect of a blur is corrected, even when the processing target image captured by the first imaging system **101** includes the effect of a blur caused by defocus. As a result, the depth of field of the first imaging system **101** can be substantially extended. Further, since the training image group **32**G and the true image **36** are created in advance based on the predetermined subject image **30** captured by the given imaging system **104**, the trained model **120** trained by machine learning in advance can be used when a subject associated with the processing target image is a subject of which image is captured by the first imaging system **101** for the first time. Further, the defocus simulation processing (step S**200**) is performed for the region on the optical axis of the first imaging system **101** and the regions other than on the optical axis in each training image **32**, based on the transfer function or the point spread function on the optical axis, thereby reducing the volume of information necessary for the defocus simulation processing (step S**200**). As a result, the trained model **120** can be created with an appropriate scale of the neural network NN necessary for machine learning. This facilitates implementation of the trained model **120** in the information processing system **100**.

[0133] The technique of the present embodiment can also be implemented as the trained model **120**. In other words, the trained model **120** in the present embodiment is used by the information processing system **100** including the memory section **110** that stores the trained model **120**, the input section **140**, the processing section **130**, and the output section **150**, and is trained by machine learning using a data set including the training image group **32**G and the true image **36**. The training image group **32**G includes a plurality of training images **32** generated by performing the defocus simulation processing that simulates the effect of a blur caused by defocus of the first imaging system **101**, for the predetermined subject image **30** in which the given imaging system **104** is focused on a predetermined subject of which image is captured by the given imaging system

104, based on the transfer function or the point spread function of the first imaging system **101** at a plurality of object distances. The defocus simulation processing is performed for the region on the optical axis of the first imaging system **101** and the regions other than on the optical axis in each training image **32** of a plurality of training images **32**, based on the transfer function or the point spread function on the optical axis. The true image **36** is an image generated by performing the best focus simulation processing that simulates, for the predetermined subject image **30**, a state in which the first imaging system **101** is focused, based on the transfer function or the point spread function at an object distance at which the first imaging system **101** is focused, or the predetermined subject image **30** itself. The trained model **120** is trained by machine learning so that each training image **32** is the true image **36**. The input section **140** inputs the processing target image, which is an image captured by the first imaging system **101**, to the trained model **120**. The processing section **130** performs the correction processing of correcting a blur caused by defocus of the first imaging system **101** in the processing target image, using the trained model **120**. The output section **150** outputs a corrected image produced by the correction processing. In this way, effects similar to those described above can be achieved.

[0134] The technique of the present embodiment can also be implemented as an information processing method. In other words, the information processing method according to the present embodiment corrects a blur caused by defocus of the first imaging system **101** in the processing target image which is an image captured by the first imaging system **101**, using the trained model **120** trained by machine learning with a data set including the training image group **32**G and the true image **36**. The training image group **32**G includes a plurality of training images **32** generated by performing the defocus simulation processing that simulates the effect of a blur caused by defocus of the first imaging system **101**, for the predetermined subject image **30** in which the given imaging system **104** is focused on a predetermined subject of which image is captured by the given imaging system **104**, based on the transfer function or the point spread function of the first imaging system **101** at a plurality of object distances. The defocus simulation processing is performed for the region on the optical axis of the first imaging system **101** and the regions other than on the optical axis in each training image **32** of a plurality of training images **32**, based on the transfer function or the point spread function on the optical axis. The true image **36** is an image generated by performing the best focus simulation processing that simulates, for the predetermined subject image **30**, a state in which the first imaging system **101** is focused, based on the transfer function or the point spread function at an object distance at which the first imaging system **101** is focused, or the predetermined subject image **30** itself. The trained model **120** is trained by machine learning so that each training image **32** is the true image **36**. In this way, effects similar to those described above can be achieved.

[0135] The technique of the present embodiment can also be implemented as an information storage medium that stores the trained model **120**. In this way, the training model **20** trained by machine learning by the training device **10** can be stored in the information storage medium. With this configuration, the information storage medium is connected to the information processing system **100**, whereby the training model **20** can be updated as the latest trained model **120**. As a result, effects similar to those described above can be achieved under a predetermined situation. The predetermined situation includes, for example, a situation in which the location of the training device **10** is distant from the location of the information processing system **100**, a situation in which data communication fails between the training device **10** and the information processing system **100**, and the like.

[0136] The technique of the present embodiment can also be implemented as the endoscope system **300**. For example, the endoscope system **300** according to the present embodiment includes a processor unit **200** including the information processing system **100** described above and an endoscopic scope **310** connected to the processor unit **200** to capture a processing target image. In this way, the endoscope system **300** including the information processing system **100** having the

aforementioned effects can be constructed.
[0137] More specifically, the endoscope system **300** may have, for example, a configuration example as illustrated in FIG. **14**. The endoscope system **300** includes the endoscopic scope **310**, an operation section **320**, a display section **330**, and the processor unit **200**. The processor unit **200** includes a storage section **210**, a control section **220**, and the information processing system **100**. The information processing system **100** in FIG. **14** further includes a storage interface **160** in addition to the configuration described above with reference to FIG. **2**. A configuration similar to that of FIG. **2** will not be further elaborated.
[0138] The endoscopic scope **310** includes an imaging device at a not-illustrated distal end thereof. The imaging device includes the first imaging system **101**. The distal end of the endoscopic scope **310** is inserted into a body cavity. The imaging device captures an image of an abdominal cavity, and captured image data is transmitted from the endoscopic scope **310** to the processor unit **200**. The operation section **320** is a device for the user to operate the endoscope system **300** and includes, for example, a button or a dial, a foot switch, or a touch panel. The display section **330** is a device that displays an image captured by the endoscopic scope **310**. The display section **330** is, for example, a liquid crystal display but may be hardware integrated with the operation section **320**, such as a touch panel.
[0139] The processor unit **200** performs control in the endoscope system **300** and processing such as image processing. For example, the control section **220** performs switching of a mode of the endoscope system **300**, a zoom operation, switching of display, or the like, based on information input from the operation section **320**, whereby the function as the processor unit **200** is implemented. The storage section **210** records an image captured by the endoscopic scope **310**. The storage section **210** is, for example, a semiconductor memory, a hard disk drive, an optical disk drive, or the like.
[0140] In the configuration example illustrated in FIG. **14**, a connector connected to a cable of the endoscopic scope **310** or an interface circuit or the like that receives imaging data is incorporated in the input section **140** to implement a function of receiving imaging data from the endoscopic scope **310**. However, the processor unit **200** may further include an interface circuit for receiving imaging data.
[0141] The storage interface **160** is an interface for accessing the storage section **210**. The storage interface **160** records image data received by the input section **140** into the storage section **210**. When replaying the recorded image data, the storage interface **160** reads the image data from the storage section **210** and transmits the image data to the processing section **130**. The processing section **130** performs the processing described above with reference to FIG. **3** on the image data from the input section **140** or the storage interface **160** as the processing target image. With this configuration, the processing section **130** outputs the corrected image via the output section **150**, and the corrected image in focus appears on the display section **330**.
[0142] The endoscope system **300** according to the present embodiment may have, for example, a configuration example illustrated in FIG. **15**. The configuration example in FIG. **15** differs from the configuration example in FIG. **14** in that the information processing system **100** and the processor unit **200** are provided separately. The information processing system **100** and the processor unit **200** may be connected by device-to-device communication such as a universal serial bus (USB), or by network communication such as a local area network (LAN) and a wide area network (WAN). The information processing system **100** includes one or more information processing devices. In a case where the information processing system **100** includes a plurality of information processing devices, the information processing system **100** may be a cloud system in which a plurality of PCs, a plurality of servers, and the like connected via a network perform parallel processing. A storage section **170** in FIG. **15** corresponds to the storage section **210** in FIG. **14**.
[0143] The processor unit **200** includes a control section **220**, an imaging data reception section **230**, an input section **240**, an output section **250**, a processing section **260**, and a display interface

270. The imaging data reception section **230** is configured with an interface circuit or the like similar to the input section **140** in FIG. **14** and receives imaging data from the endoscopic scope **310**. The processing section **260** transmits image data received by the imaging data reception section **230** to the information processing system **100** via the output section **250**. The information processing system **100** performs the processing in FIG. **3** on the received image data as the processing target image and generates a corrected image. The input section **240** receives the corrected image transmitted from the information processing system **100** via the output section **150**, and outputs the corrected image to the processing section **260**. The processing section **260** outputs the corrected image to the display section **330** via the display interface **270**. As a result, the corrected image appears on the display section **330**. The display interface **270** in FIG. **15** is configured with hardware similar to the output section **150** in FIG. **14** and implements a function similar to the output section **150** in FIG. **14**. In FIG. **15**, the input section **140** and the output section **150** of the information processing system **100** may be configured with separate interfaces, but the functions of the input section **140** and the output section **150** may be implemented by a single input/output interface. This is applicable to the input section **240** and the output section **250** of the processor unit **200**.

[0144] The technique of the present embodiment is not limited to the foregoing and various modifications may be implemented. For example, the endoscope system **300** according to the present embodiment may include the endoscopic scope **310** described above with reference to FIG. **14** and the like and an information processing system **1000** illustrated in FIG. **16**. The information processing system **1000** illustrated in FIG. **16** includes a memory section **1110** and a processing section **1130**. The memory section **1110** includes a trained model **1120** and a correction table **1400**. The memory section **1110** in FIG. **16** can be implemented by a memory similar to the memory section **110** in FIG. **1**, and the processing section **1130** can be implemented by a processor similar to the processing section **130** in FIG. **1**. The trained model **1120** and the correction table **1400** will be detailed later.

[0145] FIG. **17** is a block diagram illustrating a configuration example of the information processing system **1000** in more detail. In FIG. **17**, the information processing system **1000** further includes an input section **1140** and an output section **1150**, in addition to the memory section **1100** and the processing section **1130** described above. The input section **1140** in FIG. **17**, similar to the input section **140** in FIG. **2**, is an image data interface that receives image data as a processing target image from the first imaging system **101**. The output section **1150** in FIG. **17**, similar to the output section **1150** in FIG. **2**, is an interface that transmits a corrected image to the outside and serves the function of the output section **1150** such that output data from the trained model **1120** is the corrected image transmitted by the output section **1150**.

[0146] The processing example related to the technique of the present embodiment may be performed according to a modified flowchart illustrated in FIG. **18** instead of the flowchart illustrated in FIG. **3**. In FIG. **18**, the processing section **1130** reads the processing target image (step S**1010**) and reads the trained model (step S**1020**). Step S**1010** in FIG. **18** is processing corresponding to step S**10** in FIG. **3**, and step S**1020** in FIG. **18** is processing corresponding to step S**20** in FIG. **3**. In other words, specifically, for example, the processing section **1130** performs processing of inputting the processing target image received via the input section **1140** to the trained model **1120** read from the memory section **1110**.

[0147] The processing section **1130** thereafter performs correction processing (step S**1030**) and blur adjustment processing (step S**1032**), and outputs a blur adjustment processed image **1034** as a corrected image (step S**1040**). In this way, the corrected image in FIG. **17** differs from the corrected image in FIG. **2** in that it is the blur adjustment processed image **1034** generated by further performing blur adjustment processing (step S**1032**) for the output image **134** generated by performing correction processing (step S**1030**) for the processing target image.

[0148] The machine learning for generating the trained model **1120** in FIG. **16** will now be described. The trained model **1120** in FIG. **16** is trained by machine learning with a data set including an object distance label in addition to the training image **32** and the true image **36** described above.

[0149] More specifically, the object distance label in the present embodiment refers to an estimated object distance label **1070** and a true object distance label **1076** described later.

[0150] For example, the training model **1020** described later can be updated as the trained model **1120** by performing machine learning using the training device **10** or the like described above with reference to FIG. **4**. At least a part of the training model **1020** includes the neural network NN described above, which is similar to the training model **20** in FIG. **4**.

[0151] As illustrated in FIG. **19**, the training image **32** described above with reference to FIG. **5** and the like is input as input data to the training model **1020**. The types of training image **32** are as many as the number of object distances. In other words, as described above with reference to FIG. **10**, the first training image **32-1** is generated by performing defocus simulation processing (step S**200-1**) on the transfer function or the point spread function corresponding to the first object distance for the predetermined subject image **30**. Similarly, the Nth training image **32-**N is generated by performing defocus simulation processing (step S**200**-N) on the transfer function or the point spread function corresponding to the Nth object distance for the predetermined subject image **30**. In this way, it can be generalized that in the learning phase, machine learning is performed by inputting the Nth training image **32-**N as input data to the training model **1020**.

[0152] In FIG. **13**, the transfer function or the point spread function on the optical axis is used for the defocus simulation processing (step S**200**). This is applicable to the defocus simulation processing (step S**200**) to be used to generate the training image **32** to be input to the training model **1020** in FIG. **19**. As a result, the network scale necessary for generating the trained model **1120** in FIG. **16** and the like can be reduced.

[0153] The output image **134** is then output from the training model **1020** through computation by the neural network NN. The output image **134** is estimation result data derived by the training model **1020** from the training image **32**, which is estimation target data, and is similar to the output image **34** described above with reference to FIG. **5**.

[0154] When the training image **32** is input to the training model **1020**, the object distance corresponding to the transfer function or the point spread function used for the input training image **32** is estimated through computation by the neural network NN. As a result, the estimated object distance label **1070** associated with the estimated object distance is output together with the output image **134** described above. In other words, the estimated object distance label **1070** is a class classification result. Based on the foregoing, in the learning phase, as a result of inputting the Nth training image **32-**N as input data to the training model **1020**, the Nth output image **134-**N and the Nth estimated object distance label **1070-**N are output as output data from the training model **1020**.

[0155] FIG. **20** is a flowchart illustrating a processing example of trained model creation processing (step S**1100**) associated with the trained model **1120**. After reading a predetermined subject image (step S**1110**), the training device processing section **16** performs training data generation processing (step S**1120**). For example, the training device processing section **16** reads the predetermined subject image **30** from the training device memory section **18** and generates the training image **32** and the true image **36** using the predetermined subject image **30**, as described above with reference to FIG. **7**. The training device processing section **16** also generates the true object distance label **1076** based on the object distance corresponding to the transfer function or the point spread function associated with the defocus simulation processing (step S**200**) used to generate the training image **32**.

[0156] In FIG. **10** and FIG. **11**, the true image **36** is an image generated by performing the best focus simulation processing (step S**300**) for the predetermined subject image **30** or the predetermined subject image **30** itself. This is applicable to the true image **36** generated by step S**1120** in FIG. **20**.

[0157] The training device processing section **16** then performs first correction learning processing (step S**1131**) and second correction learning processing (step S**1132**) and terminates the flow. For example, the training device processing section **16** performs processing of reading the training model **1020** from the training device memory section **18** and inputting to the training model **1020** the input data with the training image **32** generated in the training data generation processing (step S**1120**). As a result, the output image **134** and the estimated object distance label **1070** are output as output data from the training model **1020** as described above with reference to FIG. **19**. The training device processing section **16** then performs machine learning based on the output image **134** and the true image **36** as the first correction learning processing (step S**1131**). The first correction learning processing (step S**1131**) is similar to the correction learning processing (step S**130**) described above with reference to FIG. **7** and FIG. **8**, and detailed description and illustration are omitted.

[0158] The training device processing section **16** also performs machine learning processing as the second correction learning processing (step S**1132**), based on the estimated object distance label **1070** and the true object distance label **1076** generated in step S**1120**. For example, the training device processing section **16** changes network parameters of the neural network NN so that the first estimated object distance label **1070-1** to the Nth estimated object distance label **1070**-N are the true object distance labels **1076**, as illustrated in FIG. **21**. The meaning of changing the network parameters of the neural network NN here are described above with reference to FIG. **7**.

[0159] More specifically, for example, it is assumed that the training image **32** obtained by performing the defocus simulation processing (step S**200**) for the predetermined subject image **30** using the transfer function or the point spread function at the Mth object distance is input to the training model **1020**. In this case, the true object distance label **1076** corresponding to the Mth object distance is generated by step S**1120**.

[0160] The estimated object distance label **1070** output from the training model **1020** is then classified into a class according to the estimated object distance. For example, if the object distance associated with the estimated object distance label **1070** output from the training model **1020** is the first object distance, it is classified as class **1**. Similarly, if the object distance associated with the estimated object distance label **1070** output from the training model **1020** is the Nth object distance, it is classified as class N. The same processing is repeated to construct, for example, a class classification table indicated by H**20** in FIG. **22**. In the table indicated by H**20**, a probability PR(1) is the probability that the estimated object distance associated with the estimated object distance label **1070** is the first object distance. Similarly, a probability PR(N) is the probability that the estimated object distance associated with the estimated object distance label **1070** is the Nth object distance.

[0161] Then, in the second correction learning processing (step S**1132**), the processing of calculating an error function based on the table indicated by H**20** and a table indicated by H**30** and the processing of updating a weighting factor of the neural network NN so that the error function is reduced are performed. The table indicated by H**30** is a table in which the probability that the estimated object distance associated with the estimated object distance label **1070** is the Mth object distance associated with the true object distance label **1076** is 1, and the probability that the estimated object distance associated with the estimated object distance label **1070** is an object distance other than the Mth object distance is 0.

[0162] Through the optimization of the parameters of the neural network NN in this manner, the training model **1020** is updated as the trained model **1120**. Inference using the updated trained model **1120** is then performed as illustrated in FIG. **23**. In other words, when a processing target image is input to the input layer of the neural network NN included in the trained model **1120**, computation is performed by the intermediate layer. Based on the computation result by the intermediate layer, the output image **134** is output from a first output layer and the estimated object distance label **1070** is output from a second output layer. In other words, the trained model **1120** in

FIG. **23** is constructed with one neural network NN having an output layer split into the first output layer and the second output layer, and the output image **134** and the estimated object distance label **1070** are output from the one neural network NN.
[0163] The blur adjustment processing (step S**1032**) in FIG. **18** will be described in more detail using the flowchart in FIG. **24**. The processing section **1130** performs processing of acquiring a characteristic filter from the correction table **1400** based on the estimated object distance (step S**1033**) and performs processing of adjusting a blur of the output image **134** based on the acquired characteristic filter (step S**1034**).
[0164] The correction table **1400** is a table in which object distances are associated with filter characteristics for blur correction. More specifically, for example, as indicated by H**10** in FIG. **25**, the table is a set of correction factors according to combinations of the object distance associated with the estimated object distance label **1070** output from the trained model **1120** and the frequency at image height perpendicular to the optical axis. The "object distance" in a row of the correction table **1400** is a simplified representation of the object distance associated with the estimated object distance label **1070** output from the trained model **1120**.
[0165] For example, in step S**1033**, the processing section **1130** searches the correction table **1400** for correction factor data associated with the same object distance as the object distance of the estimated object distance label **1070** generated in the correction processing (step S**1030**). Then, in step S**1034**, the processing section **1130** adjusts a blur for the output image **134** using the retrieved correction factor. For example, if the processing section **1130** retrieves correction factor data in a row indicated by H**11**, conversion into a frequency response function indicated by H**12** may be performed based on the retrieved correction factor data. In this case, the frequency response function indicated by H**12** corresponds to the characteristic filter in step S**1033**. Then, in step S**1034**, the processing section **1130** performs predetermined filter processing for the output image **134** using the frequency response function indicated by H**12**. The characteristic filter associated with step S**1033** is not limited to the frequency response function, and details will be described later with reference to FIG. **57** and FIG. **58**.
[0166] The frequency on the horizontal axis of the graph indicated by H**12** in FIG. **25** is a simplified representation of a normalized frequency. In other words, the horizontal axis of the graph indicated by H**12** in FIG. **25** is the same as the horizontal axes of the graphs in FIG. **26** and FIG. **27** described later.
[0167] In this way, the blur adjustment processed image **1034**, which is an image obtained by the processing section **1130** performing blur adjustment processing (step S**1032**) for the output image **134**, is output as a corrected image from the output section **1150** in step S**1040** in FIG. **18**. The blur adjustment processed image **1034** is output to a predetermined display device. However, the output destination is not limited to this and the corrected image may be output to a storage device of an external device or the like. Specifically, the predetermined display device is, for example, the display section **330** illustrated in FIG. **14** and the like. This is applicable to the information processing system **100** described above with reference to FIG. **2**.
[0168] Based on the foregoing, the information processing system **1000** according to the present embodiment includes the memory section **1110** and the processing section **1130**. The memory section **1110** stores the trained model **1120** trained by machine learning with a data set including the training image **32**, the true image **36**, and the object distance label, and the correction table **1400** in which an object distance which is a distance between an imaging system and a subject is associated with a filter characteristic for blur correction. The training image **32** is generated by performing the defocus simulation processing that simulates the effect of a blur caused by defocus of the first imaging system **101**, for the predetermined subject image **30** in focus captured by the given imaging system **104**, based on the transfer function or the point spread function of the first imaging system **101** at a predetermined object distance. The defocus simulation processing is performed for the region on the optical axis and the regions other than on the optical axis of the

first imaging system **101** in the training image, based on the transfer function or the point spread function on the optical axis. The true image **36** is an image generated by performing the best focus simulation processing that simulates, for the predetermined subject image **30**, a state in which the first imaging system **101** is focused, based on the transfer function or the point spread function at an object distance at which the first imaging system **101** is focused, or the predetermined subject image **30** itself. The trained model **1120** is trained by machine learning so that the training image **32** is the true image **36**, and trained by machine learning by applying, as the object distance label, the object distance of the transfer function or the point spread function of the first imaging system **101** used in the defocus simulation processing. The processing section **1130** uses the trained model **1120** to generate the output image **134** in which a blur of the processing target image which is an image captured by the first imaging system **101** is corrected, and estimates an object distance of the processing target image. The processing section **1130** acquires a filter characteristic associated with the estimated object distance from the correction table **1400**, and performs blur adjustment processing for the output image **134** using the acquired filter characteristic.

[0169] In this way, the information processing system **1000** according to the present embodiment, which includes the memory section **110** that stores the trained model **1120** and the correction table **1400** and the processing section **130**, can output a corrected image in which the effect of a blur is corrected more appropriately from the processing target image. As a result, the depth of field of the first imaging system **101** can be substantially extended. Further, the defocus simulation processing (step S**200**) is performed for the region on the optical axis of the first imaging system **101** and the regions other than on the optical axis in each training image **32**, based on the transfer function or the point spread function on the optical axis, thereby reducing the volume of information necessary for the defocus simulation processing (step S**200**). As a result, the trained model **1120** can be created with an appropriate scale of the neural network NN necessary for machine learning. This facilitates implementation of the trained model **1120** in the information processing system **1000**.

[0170] The technique described above can be implemented as the endoscope system **300** described above with reference to FIG. **14** or FIG. **15**. In other words, the endoscope system **300** according to the present embodiment includes the information processing system **1000** described above and an endoscopic scope **310** that captures a processing target image. In this way, effects similar to those described above can be achieved.

[0171] The technique of the present embodiment may also be implemented as an image processing method. In other words, the technique of the present embodiment relates to an image processing method using the trained model **1120** trained by machine learning with a data set including the training image **32**, the true image **36**, and the object distance label, and the correction table **1400** in which an object distance which is a distance between an imaging system and a subject is associated with a filter characteristic for blur correction. The image processing method includes a step of using the trained model **1120** to generate the output image **134** in which a blur of the processing target image which is an image captured by the first imaging system **101** is corrected, and a step of estimating an object distance of the processing target image (step S**1030**). The image processing method further includes a step of acquiring a filter characteristic associated with the estimated object distance from the correction table **1400**, and performing blur adjustment processing for the output image **134** using the acquired filter characteristic (step S**1032**). The training image **32** is generated by performing the defocus simulation processing that simulates the effect of a blur caused by defocus of the first imaging system **101**, for the predetermined subject image **30** in focus captured by the given imaging system **104**, based on the transfer function or the point spread function of the first imaging system **101** at a predetermined object distance. The defocus simulation processing is performed for the region on the optical axis of the first imaging system **101** and the regions other than on the optical axis in the training image **32**, based on the transfer function or the point spread function on the optical axis. The true image **36** is an image generated by performing the best focus simulation processing that simulates, for the predetermined subject image **30**, a state in which the first imaging system **101** is focused, based on the transfer function or the point spread function at an object distance at which the first imaging system **101** is focused, or the predetermined subject image **30** itself. The trained model **1120** is trained by machine learning so that the training image **32** is the true image **36**, and trained by machine learning by applying, as the object distance label, the object distance of the transfer function or the point spread function of the first imaging system **101** used in the defocus simulation processing. In this way, effects similar to those described above can be achieved.

[0172] The technique described above may also be implemented as a program. In other words, the program according to the present embodiment causes a computer to execute the image processing method described above. The technique of the present embodiment may be implemented as a non-transitory information storage medium including the program described above. In this way, effects similar to those described above can be achieved.

[0173] In the information processing system **1000** according to the present embodiment, the trained model **1120** may be configured with one neural network NN, and the neural network NN may include an input layer to which a processing target image is input, an intermediate layer that performs computation for an output from the input layer, a first output layer that generates the output image **134** from an output from the intermediate layer, and a second output layer that estimates an object distance from an output from the intermediate layer. In this way, increase of the network scale can be prevented.

[0174] The technique of the present embodiment is not limited to the foregoing and various modifications may be implemented. For example, each object distance included in the optical system information **40** may be determined based on a difference in the corresponding MTF. For example, it is assumed that the training image group **32**G includes the first training image **32-1** that undergoes step S**200-1** based on the transfer function or the point spread function at the first object distance, and the second training image **32-2** that undergoes step S**200-2** based on the transfer function or the point spread function at the second object distance. Further, it is assumed that the first object distance is an object distance with a larger amount of defocus, compared with the second object distance. In this case, when a spatial frequency dependence of MTF is qualitatively illustrated, the MTF based on the second object distance is as indicated by A**0** in FIG. **26**, and the MTF based on the first object distance is as indicated by A**1**. Then, for example, when a predetermined spatial frequency indicated by B**0** is determined, the difference of MTF is determined as indicated by C**0**. The first object distance and the second object distance are then determined so that the difference of MTF indicated by C**0** is smaller than a predetermined value.

[0175] The difference of MTF here is a difference of MTF between adjacent object distances. For example, it is assumed that the training image group **32**G includes the first training image **32-1**, the second training image **32-2**, and the third training image **32-3**. Further, it is assumed that the amount of defocus is larger in the order of the first object distance, the second object distance, and the third object distance. In this case, A**10** in FIG. **27** indicates the frequency characteristic of the MTF at the third object distance, A**11** indicates the frequency characteristic of the MTF at the second object distance, and A**12** indicates the frequency characteristic of the MTF at the first object distance. Then, it is assumed that at a predetermined frequency indicated by B**0**, both of the difference between the MTF of A**10** and the MTF of A**11** as indicated by C**10** and the difference between the MTF of A**11** and the MTF of A**12** as indicated by C**11** are lower than a predetermined value. In other words, at the predetermined frequency indicated by B**0**, the difference between the MTF of A**10** and the MTF of A**12** is not considered as the predetermined value. Based on the foregoing, in the information processing system **100** according to the present embodiment, the object distance is set so that the difference in value of MTF between adjacent object distances is equal to or less than a predetermined value, at a predetermined spatial frequency of the MTF of the first imaging system **101**. In this way, an appropriate combination of data sets in machine learning can be set. As described above, the trained model **120** trained by machine learning performs correction processing (step S**30**) so that both of the first training image **32-1** and the second training image **32-2** can be corrected to the true image **36**. In addition, in order to correct the processing target image captured at an object distance between the first object distance and the second object distance to the true image **36** through the correction processing (step S**30**), it is preferable that the difference between the effects of a blur added to the first training image **32-1** and the second training image **32-2** is within a certain range. In this respect, the technique of the present embodiment can be applied to generate an appropriate training image group **32**G, because the object distance of each training image is defined based on the MTF that indicates the effect of a blur simulated for the predetermined subject image **30**. As a result, an appropriate data set can be set in machine learning.

[0176] Further, the optical system information **40** may include an object distance in the best focus condition of the first imaging system **101**. The object distance in the best focus condition is specifically, for example, the distance indicated by D**3** in FIG. **9**. For example, the training device processing section **16** may generate the true image **36** by performing the best focus simulation processing (step S**300**) for the predetermined subject image **30** using the transfer function or the point spread function using the object distance in the best focus condition. In other words, in the information processing system **100** according to the present embodiment, the object distance that achieves focus is the object distance in the best focus condition. In this way, an appropriate true image **36** can be generated.

[0177] In the present embodiment, it is assumed that the transfer function or the point spread function based on the object distance has one-to-one correspondence with the training image **32**. More specifically, for example, in the defocus simulation processing (step S**200**), it is assumed that processing of generating the third training image **32-3** is not performed using both of the transfer function or the point spread function with the first object distance and the transfer function or the point spread function with the second object distance for one predetermined subject image **30**. In other words, in the information processing system **100** according to the present embodiment, each training image **32** is an image generated by performing the defocus simulation processing (step S**200**) for the predetermined subject image **30** based on the transfer function or the point spread function at any one object distance of a plurality of object distances. In this way, the relationship between the training images **32** in the training image group **32**G can be clarified.

[0178] In a common optical system, as the spatial frequency increases, the MTF decreases and changes with periodicity. Since the MTF is an absolute value, the MTF is displayed in a folding manner in a high spatial frequency region indicated by B**1** in FIG. **27**. Thus, in the high spatial frequency region, it is impossible to uniquely determine which object distance one MTF corresponds to. For example, the MTF at an object distance shorter than the object distance at the near point of the target extended depth of field as indicated by P**2** in FIG. **9** may be zero in the spatial frequency indicated by B**0**. For example, supposing that A**12** in FIG. **27** is the MTF at the object distance at the near point of the target extended depth of field, a spatial frequency lower than the lowest spatial frequency at which folding occurs may be the spatial frequency indicated by B**0**. This is because the transfer function or the point spread function at an object distance outside the target depth of field is not used in the machine learning in the present embodiment in the first place. The target extended depth of field here does not exhibit a constant value, as described above. Based on the foregoing, in the information processing system **100** according to the present embodiment, the processing section **130** uses the trained model **120** to correct a blur caused by defocus of the first imaging system **101** for the processing target image, thereby estimating an image in which the depth of field of the first imaging system **101** is extended to the target extended depth of field wider than the depth of field. Further, the predetermined spatial frequency is a spatial frequency lower than the lowest spatial frequency at which the value of MTF at the near point of the target extended depth of field is zero. In this way, the range of the predetermined spatial frequency necessary for associating the spatial frequency with MTF in a one-to-one correspondence can be determined as appropriate.

[0179] More specifically, it is desired that the predetermined spatial frequency indicated by B**0** is, for example, 0.1 as a normalized frequency. In other words, in the information processing system **100** according to the present embodiment, the predetermined spatial frequency is a spatial frequency that is ⅕ of the Nyquist frequency of an image sensor of the first imaging system **101**. In this way, the spatial frequency can be associated with the MTF in a one-to-one correspondence for many optical systems. As a result, the technique of the present embodiment can be applied to the processing target image captured by various kinds of optical systems.

[0180] Further, the optical system information **40** in the present embodiment may be a combination of an object distance inside the depth of field and an object distance outside the depth of field. Specifically, for example, the optical system information **40** may include the first object distance outside the depth of field indicated by D**1** in FIG. **9** and the second object distance indicated by D**2**. In other words, in the information processing system **100** according to the present embodiment, the first object distance of a plurality of object distances is an object distance outside the depth of field, and the second object distance of a plurality of object distances is an object distance inside the depth of field. This configuration results in data sets in which the first training image **32-1** that simulates the effect of a blur to a large degree by the defocus simulation processing (step S**200**) and the second training image **32-2** that simulates the effect of a blur to a small degree are combined with the true image **36**. As a result, the trained model **120** trained by machine learning with these data sets can correct the processing target image having the effect of a blur in a wide range, through the correction processing (step S**30**).

[0181] Further, the predetermined value may be determined based on the number of training images **32** that constitute the training image group **32**G. For example, in FIG. **26**, it is assumed that the MTF indicated by A**0** is the MTF at the object distance corresponding to the best focus condition, and the MTF indicated by A**1** is the MTF at the object distance corresponding to the near point of the target depth of field. In this case, for example, when the spatial frequency is determined as the spatial frequency indicated by B**0**, the range of MTF is uniquely determined such that the range indicated by C**0** is the largest. Then, a value obtained by dividing the range indicated by C**0** based on a desired number of training images **32** is determined as the predetermined value. Based on the foregoing, in the information processing system **100** according to the present embodiment, the predetermined value is determined based on the number of object distances that can be set to two or more. In this way, the number of data sets necessary for machine learning can be determined in consideration of the load of machine learning.

[0182] As described above, since the range of MTF is uniquely determined by fixing the spatial frequency, the predetermined value may be determined in advance and the number of training images **32** may be determined based on the predetermined value. The user can determine a policy of machine learning depending on a situation.

[0183] It is desirable that the predetermined value is equal to or less than 0.2. In other words, in the information processing system **100** according to the present embodiment, the predetermined value is set to be equal to or less than 0.2. In a common optical system, when the aforementioned spatial frequency is determined in a desirable range, a possible range of MTF is presumably about 0.2. Thus, for example, when the predetermined value is set to 0.2, the number of training images **32** that constitute the training image group **32**G is two. In this case, presumably, the first object distance is an object distance outside the depth of field, and the second object distance is an object distance inside the depth of field.

[0184] Further, it is desirable that the predetermined value is equal to or less than 0.1. In other words, in the information processing system **100** according to the present embodiment, the predetermined value is set to be equal to or less than 0.1. Further, it is desirable that the predetermined value is equal to or less than 0.05. In other words, in the information processing system **100** according to the present embodiment, the predetermined value is set to be equal to or less than 0.05. In this way, the number of training images **32** that constitute the training image group **32**G can be increased. With this configuration, when a processing target image captured at an object distance other than an object distance not used in machine learning is input, the trained model **120** is more likely to output a corrected image from which the effect of a blur is appropriately removed. In other words, the accuracy of the correction processing (step S**30**) by the trained model **120** can be improved more. If the number of training images **32** that constitute the training image group **32**G increases, the processing load of machine learning increases. Thus, an appropriate number of training images **32** that constitute the training image group **32**G is determined as appropriate depending on a situation.

[0185] A specific technique for the training device processing section **16** to perform the defocus simulation processing (step S**200**) and the like using the point spread function will now be described. For example, in a case where the first training image **32-1** is generated by step S**200-1**, as illustrated in FIG. **28**, the training device processing section **16** performs convolution computation processing for the predetermined subject image **30** using the PSF at the first object distance of the first imaging system **101**. The convolution can also be referred to as convolution integral. The PSF at the first object distance here is the PSF of the region indicated by FC**22-1** in FIG. **12**. In other words, in the case of the technique in FIG. **28**, the convolution computation processing of the PSF corresponds to step S**210** in FIG. **13**. Similarly, in a case where the Nth training image **32-**N is generated by step S**200-**N, the training device processing section **16** performs convolution computation processing for the predetermined subject image **30** using the PSF at the Nth object distance of the first imaging system **101**. The defocus simulation processing based on the convolution computation processing of the PSF can be called step S**200-**A. Based on the foregoing, in the information processing system **100** according to the present embodiment, the defocus simulation processing (step S**200**) is processing of performing convolution computation of the PSF at each of object distances of the first imaging system **101** for the predetermined subject image **30**. In this way, the trained model **120** trained by machine learning with a data set of the training images **32** using the PSF and the true image **36** can be generated.

[0186] A specific technique for the training device processing section **16** to perform the defocus simulation processing (step S**200**) using the transfer function will now be described. For example, in a case where the first training image **32-1** is generated, as illustrated in FIG. **29**, the training device processing section **16** performs processing of performing Fourier transform of the predetermined subject image **30**, processing of multiplying a frequency characteristic which is the result of the Fourier transform by the OTF at the first object distance of the first imaging system **101**, and processing of performing inverse Fourier transform of the multiplied frequency characteristic. The OTF at the first object distance here is the OTF of the region indicated by FC**22-1** in FIG. **12**. In other words, in the case of the technique in FIG. **29**, the multiplication by the OTF corresponds to step S**210** in FIG. **13**. Similarly, in a case where the Nth training image **32-**N is generated by step S**200-**N, the training device processing section **16** performs processing of performing Fourier transform of the predetermined subject image **30**, processing of multiplying a frequency characteristic which is the result of the Fourier transform by the OTF at the Nth object distance of the first imaging system **101**, and processing of performing inverse Fourier transform of the multiplied frequency characteristic. The defocus simulation processing based on the multiplication by the OTF can be called step S**200-**B. Based on the foregoing, in the information processing system **100** according to the present embodiment, the defocus simulation processing (step S**200**) is processing of performing Fourier transform of the predetermined subject image **30**, multiplying the frequency characteristic of the predetermined subject image **30** which is the result of the Fourier transform by the OTF at each object distance of the first imaging system **101**, and performing inverse Fourier transform of the multiplied frequency characteristic. In this way, the trained model **120** trained by machine learning with a data set of the training images **32** using the OTF and the true image **36** can be generated.

[0187] Since the relationship between PSF and OTF is as described above, the computation processing result for the processing in FIG. **28** and the computation processing result for the processing in FIG. **29** are mathematically equivalent. Which of PSF and OTF is to be used in the defocus simulation processing (step S**200**) may be selected as appropriate by the user.
[0188] Similarly, the training device processing section **16** may perform the best focus simulation processing (step S**300**) using the point spread function. For example, as illustrated in FIG. **30**, the training device processing section **16** generates the true image **36** by performing convolution computation processing for the predetermined subject image **30**, using the PSF at the object distance at which the first imaging system **101** is focused. The best focus simulation processing based on the convolution computation processing of the PSF can also be called step S**300**-A.
[0189] Further, the training device processing section **16** may perform the best focus simulation processing (step S**300**) using the transfer function. For example, as illustrated in FIG. **31**, the training device processing section **16** generates the true image **36** by performing processing of performing Fourier transform of the predetermined subject image **30**, processing of multiplying a frequency characteristic which is the result of the Fourier transform by the OTF at the object distance at which the first imaging system **101** is focused, and processing of performing inverse Fourier transform of the multiplied frequency characteristic. The best focus simulation processing based on the multiplication by the OTF can also be called step S**300**-B.
[0190] In the following description, an example in which the training image **32** and the true image **36** are generated using the technique using the PSF will be described, but it is not intended to preclude the technique using the OTF from being applied.
[0191] For example, the first imaging system **101** in the present embodiment may have a retrofocus lens configuration. The retrofocus configuration is also called inverted telephoto configuration. For example, the retrofocus lens configuration can be implemented, for example, by disposing a lens with a negative refractive power and a lens with a positive refractive power from the subject side. In the following description, a lens group on the subject side is referred to as front lens group, and a lens group on the image side is referred to as rear lens group.
[0192] Various known configurations can be employed as a specific retrofocus lens configuration. For example, an optical system illustrated in FIG. **32** includes, in order from the subject side, a front lens group indicated by G**1**, an aperture stop indicated by S**1**, a rear lens group indicated by G**2**, and a cover glass indicated by CG**1**. In FIG. **32**, a space between lenses included in the optical system is not accurately illustrated for convenience of explanation. For example, in FIG. **32**, a positive lens indicated by L**6** and the cover glass indicated by CG**1** are actually cemented but are depicted as being spaced apart from each other for the sake of convenience. This is applicable to FIG. **33** and FIG. **35** described later.
[0193] In FIG. **32**, the front lens group indicated by G**1** includes an objective-side negative lens indicated by L**1** and a positive lens indicated by L**2** and has a negative refractive power as a whole. The rear lens group indicated by G**2** includes a positive lens indicated by L**3**, a cemented lens including a positive lens indicated by L**4** and a negative lens indicated by L**5**, and a positive lens indicated by L**6**, and has a positive refractive power as a whole.
[0194] The front lens group or the rear lens group may include a plurality of lens groups. For example, in the first imaging system **101** illustrated in FIG. **33**, a lens group indicated by G**11** functions as a front lens group, and a lens group indicated by G**12** and a lens group indicated by G**13** function as a rear lens group. For example, the lens group indicated by G**11** includes, in order from the subject side, a planoconcave lens having a concave surface facing the image side as indicated by L**11** and a negative meniscus lens as indicated by L**12** and has a negative refractive power as a whole.
[0195] For example, the lens group indicated by G**12** includes a subject-side positive lens indicated by L**13** and an image-side positive lens indicated by L**14**. An aperture stop indicated by S**11** may be further disposed between the lens indicated by L**13** and the lens indicated by L**14**. In this way, the

optical system is configured such that the refractive power is symmetric with respect to the aperture stop, thereby favorably correcting coma and astigmatism.

[0196] The lens group indicated by G**13** has a positive refractive power as a whole. Further, the lens group indicated by G**13** may include a cemented lens including a positive lens indicated by L**15** and a negative lens indicated by L**16**. This configuration can favorably correct spherical aberration and coma. Further, the lens group indicated by G**13** may further include a planoconvex lens indicated by L**17**. This configuration can ensure a wide field of view. The planoconvex lens indicated by L**17** and the cover glass indicated by CG**11** are depicted as being spaced apart from each other in FIG. **33** but actually they are cemented. The cover glass indicated by CG**11** is provided in a not-illustrated image sensor, and the planoconvex lens indicated by L**17** is used for positioning of the image sensor.

[0197] Further, for example, the first imaging system **101** may further include a parallel plate. The parallel plate is also called a filter. The parallel plate is disposed, for example, at a position F**1** in FIG. **32** or a position F**11** in FIG. **33** but may be disposed at another position. The parallel plate is used, for example, for the purpose of adjusting the position of image point.

[0198] In the first imaging system **101** including the retrofocus lens configuration as described above, it is desirable that the amount of distortion at a maximum angle of view is equal to or less than −30%. Specifically, for example, it is assumed that an image of a subject indicated by E**1** in FIG. **34** is captured as an image indicated by E**2** in FIG. **34** by the first imaging system **101**. In this case, the value of the amount of distortion (%) at the maximum angle of view can be expressed by (AD−PD)/PD×100 using a length indicated by PD of the subject indicated by E**1** and a length indicated by AD of the image indicated by E**2**. It is desirable that the value is a negative value smaller than −30. Based on the foregoing, in the information processing system **100** according to the present embodiment, the first imaging system **101** has a retrofocus lens configuration and the amount of distortion at the maximum angle of view is equal to or less than −30%. In this way, the magnification in a periphery is smaller than that of an image center, so that the transfer function or the point spread function of a region other than on the optical axis can be reduced. Further, the difference between the transfer function or the point spread function of the region on the optical axis and the transfer function or the point spread function of a region other than on the optical axis can be reduced. As a result, the training image **32** with a more accurate result of simulation of the effect of a blur can be generated.

[0199] The front lens group or the rear lens group may be configured with a single lens. For example, the first imaging system **101** illustrated in FIG. **35** includes a lens group indicated by G**21**, a lens group indicated by G**22**, an aperture stop indicated by S**21**, a lens group indicated by G**23**, and a cover glass indicated by CG**21**. The lens group indicated by G**21** includes a single negative lens indicated by L**21** and has a negative refractive power. In other words, the lens group indicated by G**21** functions as a part of the front lens group. The lens group indicated by G**23** includes a positive lens indicated by L**23**, a cemented lens including a positive lens indicated by L**24** and a negative lens indicated by L**25**, and a positive lens indicated by L**26**, and has a positive refractive power as a whole. In other words, the lens group indicated by G**23** functions as a rear lens group.

[0200] Further, the first imaging system **101** in the present embodiment may further include a phase modulation element. For example, a second lens group G**2** in FIG. **35** includes a positive lens indicated by L**22**, an aperture stop indicated by S**21**, and a phase modulation element indicated by PM. The phase modulation element indicated by PM is disposed at the pupil of the first imaging system **101**. The phase modulation element indicated by PM is an element that employs wavefront coding (WFC) and, for example, has a phase modulation surface indicated by PMS. The wave coding is a known technique used in extended depth of field (EDOF) and will not be further elaborated here.

[0201] In FIG. **35**, the phase modulation surface indicated by PMS is illustrated so as to be

represented by a predetermined cubic function using a coordinate orthogonal to the optical axis, but the surface shape of the phase modulation surface is not limited to this and another surface shape may be employed. Further, in FIG. **35**, the phase modulation surface is illustrated on the image side but can be disposed on the subject side to achieve a similar effect. Further, the lens group indicated by G**22** has a positive refractive power as a whole and also functions as a part of a retrofocus front lens group.
[0202] Further, with the inclusion of the phase modulation element indicated by PM, the MTF of the first imaging system **101** less changes with defocus. In other words, the inclusion of the phase modulation element acts such that the MTF of the first imaging system **101** matches with a change in object distance. More specifically, for example, the difference between the MTF of the first object distance and the MTF of the second object distance in the first imaging system **101** that includes the phase modulation element is smaller than the difference between the MTF of the first object distance and the MTF of the second object distance in the first imaging system **101** that does not include the phase modulation element.
[0203] For example, in a relationship between MTF and spatial frequency illustrated in FIG. **36**, it is assumed that A**20** is the MTF of the first imaging system **101** at an object distance that achieves focus, A**21** is the MTF at an object distance with a larger amount of defocus than that of the object distance associated with A**20**, and A**22** is the MTF at an object distance with a larger amount of defocus than that of the object distance associated with A**21**. Further, it is assumed that A**20** to A**22** are the MTF of the first imaging system **101** that does not include the phase modulation element. When the aforementioned predetermined spatial frequency indicated by B**0** is determined, the difference between the MTF of A**20** and the MTF of A**21** is a difference indicated by C**20**, and the difference between the MTF of A**21** and the MTF of A**22** is a difference indicated by C**21**. In FIG. **36**, the MTF in a frequency higher than the spatial frequency indicated by B**0** is partially not illustrated.
[0204] Here, since the phase modulation element indicated by PM is included in the first imaging system **101**, the MTF indicated by A**20** changes to the MTF indicated by A**30**, the MTF indicated by A**21** changes to the MTF indicated by A**31**, and the MTF indicated by A**22** changes to the MTF indicated by A**32**. Further, the difference in MTF indicated by C**20** is reduced as indicated by C**30**, and the difference in MTF indicated by C**21** is reduced as indicated by C**31**. Based on the foregoing, in the information processing system **100** according to the present embodiment, the first imaging system **101** further includes an optical wavefront modulation element that changes the transfer function or the point spread function. In this way, the distance necessary for machine learning can be reduced, so that the number of data sets necessary for machine learning can be reduced.
[0205] The example of the defocus simulation processing (step S**200**) and the like described above is a processing example for generating the training image **32** based on optical information of the first imaging system **101** for the predetermined subject image **30** captured by the given imaging system **104**. The technique of the present embodiment is not limited thereto. For example, the training device processing section **16** may perform the defocus simulation processing so as to further include processing that simulates removal of the effect of imaging by the given imaging system **104** from the predetermined subject image **30**.
[0206] FIG. **37** illustrates an example of the image data generation processing in a case of further including the processing that simulates, for a predetermined subject image **30-1** captured by the first imaging system **101**, removal of the effect of imaging by removal of the effect of the first imaging system **101**. The image data generation processing illustrated in FIG. **37** can also be called step S**122**. Step S**122** in FIG. **37** differs from step S**120-2** in FIG. **11** in the content of the defocus simulation processing. FIG. **37** is common to FIG. **11** in that the best focus simulation processing (step S**300**) is not performed and the true image **36** is the predetermined subject image **30-1** itself. This is because the predetermined subject image **30-1** is an image captured under the best focus

condition of the first imaging system **101** and there is no need for performing processing similar to step S**202** in the first place.

[0207] FIG. **38** illustrates an example of defocus simulation processing (step S**202-1**) in the image data generation processing (step S**122**). For example, in a case where the first training image **32-1** is generated, the training device processing section **16** performs the processing of simulating, for the predetermined subject image **30-1**, removal of the effect of the first imaging system **101** at the time of capturing the predetermined subject image **30-1** (step S**220-1**). Step S**220-1** is performed based on the transfer function or the point spread function at the object distance at which the first imaging system **101** is focused, and the transfer function or the point spread function at the first object distance of the first imaging system **101**.

[0208] More specifically, for example, the training device processing section **16** performs, for the predetermined subject image **30**, computation processing that appropriately combines computation processing of performing deconvolution of the PSF at the object distance at which the first imaging system **101** is focused and computation processing of performing convolution of the PSF at the first object distance of the first imaging system **101** (step S**200**-A). The "computation processing that appropriately combines" refers to computation processing in which one computation processing is combined with part or the whole of the other computation processing in a given order, but it is not intended to preclude one computation processing and the other computation processing from being performed separately. The computation processing that appropriately combines is determined as appropriate depending on a predetermined situation. This is applicable in the following description. The predetermined situation is, for example, the processing time required for machine learning, the processing load on processors, and the like. In other words, step S**220-1** can be performed to obtain, for example, a computation processing result that reflects both of the effect of the computation processing of performing deconvolution of the PSF at the object distance at which the first imaging system **101** is focused and the effect of the computation processing of performing convolution of the PSF at the first object distance of the first imaging system **101** (step S**200**-A), for the predetermined subject image **30-1**.

[0209] Based on the foregoing, in the information processing system **100** according to the present embodiment, the given imaging system **104** is the first imaging system **101**. The defocus simulation processing (step S**202**) further includes processing of removing the effect of the first imaging system **101** from the predetermined subject image **30-1**, based on the transfer function or the point spread function at the object distance at which the first imaging system **101** is focused, and the transfer function or the point spread function at a plurality of object distances of the first imaging system **101** (step S**212**). In this way, a more accurate training image **32** can be generated. The training image **32** and the true image **36** according to the technique illustrated in FIG. **10** and FIG. **11** have both the effect of the given imaging system **104** and the effect of the first imaging system **101** on a predetermined subject, whereas the training image **32** and the true image **36** according to the technique illustrated in FIG. **37** and FIG. **38** have only the effect of the first imaging system **101** on a predetermined subject. With this configuration, machine learning with more appropriate data sets can be performed.

[0210] Similarly, FIG. **39** illustrates an example of the image data generation processing including the processing that simulates removal of the effect of imaging by the given imaging system **104**. In FIG. **39**, a second imaging system **102** is illustrated as a representative of the given imaging system **104**. Further, it is assumed that the second imaging system **102** is an imaging system with an image sensor with a higher resolution, compared with the first imaging system **101**. Further, the image data generation processing illustrated in FIG. **39** can also be called step S**124**, and an original image for step S**124** can also be called a predetermined subject image **30-2**.

[0211] Step S**126** in FIG. **39** differs from step S**120-1** in FIG. **10** in that the defocus simulation processing (step S**204**) and the best focus simulation processing (step S**304**) are performed after image sensor information **50** is further read. The image sensor information **50** is information regarding a resolution of an image sensor included in each of the first imaging system **101** and the given imaging system **104**. In other words, in a case of the example in FIG. **39**, the image sensor information **50**, which is not depicted in FIG. **4**, is further stored in the training device memory section **18**. The image sensor information **50** is also used in computation processing in the defocus simulation processing (step S**204**) and the best focus simulation processing (step S**304**).

[0212] FIG. **40** illustrates an example of the defocus simulation processing in the image data generation processing (step S**124**) illustrated in FIG. **39**. The defocus simulation processing illustrated in FIG. **39** and FIG. **40** can also be called step S**204**. For example, in a case where the first training image **32-1** is generated, the training device processing section **16** performs, for the predetermined subject image **30-2**, computation processing that appropriately combines processing of simulating the difference between the second imaging system **102** and the first imaging system **101** (step S**230-1**), processing of reducing the predetermined subject image **30** (step S**240**), and computation processing based on the image sensor information **50** not depicted in FIG. **40**. Step S**230-1** is performed based on the transfer function or the point spread function at an object distance at which the second imaging system **102** is focused, and the transfer function or the point spread function at the first object distance of the first imaging system **101**. In other words, step S**230-1** can be performed to obtain a computation processing result that reflects both of the effect of the computation processing of performing deconvolution of the PSF at the object distance at which the second imaging system **102** is focused and the effect of the computation processing of performing convolution of the PSF at the first object distance of the first imaging system **101** (step S**200-**A), for the predetermined subject image **30-2**. Further, step S**204-1** can be performed to obtain a computation processing result that reflects the effect of the computation processing in step S**230-1**, the effect of the computation processing in step S**240**, and the effect of the computation processing based on the image sensor information **50**.

[0213] FIG. **41** illustrates an example of the best focus simulation processing illustrated in FIG. **39**. The best focus simulation processing illustrated in FIG. **39** and FIG. **41** can also be called step S**304**. For example, the training device processing section **16** performs, for the predetermined subject image **30-2**, processing that appropriately combines processing of simulating the difference between the second imaging system **102** and the first imaging system **101** (step S**330**), processing of reducing the predetermined subject image **30-2** (step S**340**), and computation processing based on the image sensor information **50** not depicted in FIG. **41**. As a result, the training device processing section **16** can generate the true image **36**. Step S**330** in FIG. **41** is performed based on the transfer function or the point spread function at the object distance at which the second imaging system **102** is focused, and the transfer function or the point spread function at the object distance at which the first imaging system **101** is focused. In other words, step S**330** can be performed to obtain, for example, a computation processing result that reflects both of the effect of the computation processing of performing deconvolution of the PSF at the object distance at which the second imaging system **102** is focused and the effect of the computation processing of performing convolution of the PSF at the distance at which the first imaging system **101** is focused (step S**300-**A), for the predetermined subject image **30-2**. Further, step S**340** in FIG. **41** is computation processing similar to step S**240** in FIG. **40**. Further, step S**304** can be performed to obtain a computation processing result that reflects the effect of the computation processing in step S**330**, the effect of the computation processing in step S**340**, and the effect of the computation processing based on the image sensor information **50**. The true image **36** may be generated by processing in which step S**330** is eliminated from the best focus simulation processing (step S**304**) in FIG. **41**. In other words, the true image **36** may be generated by performing processing corresponding to step S**340** for the predetermined subject image **30-2**. This is because if the predetermined subject image **30-2** is an image captured at the object distance at which the given imaging system **104** is focused, the true image **36** can be obtained by changing the number of pixels of the predetermined subject image **30-2** in step S**340**.

[0214] Based on the foregoing, in the information processing system **100** according to the present embodiment, the defocus simulation processing (step S**204**) further includes processing of simulating the difference between the given imaging system **104** and the first imaging system **101** (step S**230**) and processing of reducing the predetermined subject image **30-2** (step S**240**). The true image **36** is an image generated by performing the best focus simulation processing (step S**304**) or an image generated by performing the processing that reduces the predetermined subject image **30-2**. The processing of simulating the difference between the given imaging system **104** and the first imaging system **101** (step S**230**) in the defocus simulation processing (step S**204**) is based on the transfer function or the point spread function at the object distance at which the given imaging system **104** is focused, and the transfer function or the point spread function at a plurality of object distances of the first imaging system **101**. The best focus simulation processing (step S**304**) further includes processing of simulating the difference between the given imaging system **104** and the first imaging system **101** (step S**330**) and processing of reducing the predetermined subject image **30-2** (step S**340**). The processing of simulating the difference between the given imaging system **104** and the first imaging system **101** (step S**330**) in the best focus simulation processing (step S**304**) is based on the transfer function or the point spread function at the object distance at which the given imaging system **104** is focused, and the transfer function or the point spread function at the object distance at which the first imaging system **101** is focused.

[0215] Further, the technique of the present embodiment can also be applied to a case where the given imaging system **104** and the first imaging system **101** employ different imaging methods. For example, as illustrated in FIG. **42**, it is assumed that the first imaging system **101** includes a simultaneous-type image sensor **106**. Further, as illustrated in FIG. **43**, it is assumed that the given imaging system **104** includes a monochrome image sensor **108**. Referring to FIG. **43**, a technique of image data generation processing in this case will be described. The image data generation processing in FIG. **43** can also be called step S**126**, and an original image for step S**126** can also be called a predetermined subject image **30-3**. FIG. **43** differs from FIG. **39** in the contents of the defocus simulation processing (step S**206**) and the best focus simulation processing (step S**306**) and in that color shift determination processing (step S**190**) is performed before steps S**206** and S**306** are performed. In FIG. **43**, the second imaging system **102** is illustrated as a representative of the given imaging system **104**, which is the same as in the example in FIG. **39**. Further, the color shift determination processing (S**190**) is, for example, processing of comparing a coloring amount in a periphery of a saturated portion or the like in the predetermined subject image **30-3** with a predetermined threshold. The color shift is a shift that occurs among an R image, a G image, and a B image, for example, due to a difference in imaging timing when an image of a subject is captured using the monochrome image sensor **108**. The color shift does not occur in a processing target image captured by the simultaneous-type image sensor **106**. Further, the coloring amount in a periphery of a saturated portion or the like in the predetermined subject image **30-3** is a coloring amount that occurs due to the color shift in a periphery of an area appearing white in the predetermined subject image **30-3**. In other words, steps S**206** and S**306** in FIG. **43** use the predetermined subject image **30-3** in which the coloring amount in a periphery of a saturated portion or the like is determined to be equal to or less than a predetermined threshold in step S**190**. Thus, the training image **32** in which the effect of the color shift is reduced can be generated by performing step S**206**. Similarly, the true image **36** in which the effect of the color shift is reduced can be generated by performing step S**306**. With this configuration, in the case where the given imaging system **104** and the first imaging system **101** employ different imaging methods, an appropriate data set including the training images **32** and the true image **36** can be generated.

[0216] FIG. **44** illustrates an example of the defocus simulation processing in the image data generation processing (step S**126**) illustrated in FIG. **43**. The defocus simulation processing illustrated in FIG. **43** and FIG. **44** can also be called step S**206**. FIG. **44** differs from FIG. **40** in that it further includes processing of generating a mosaic image from the predetermined subject image

30-3 (step S**250**) and processing of demosaicing the mosaic image (step S**252**). For example, in a case where the first training image **32-1** is generated, the training device processing section **16** performs, for the predetermined subject image **30-3**, computation processing that appropriately combines the above step S**230-1**, the above step S**240**, step S**250**, step S**252**, and computation processing based on the image sensor information **50** not depicted in FIG. **44**. In other words, step S**206-1** can be performed to obtain a computation processing result that reflects the effect of the computation processing in step S**230-1**, the effect of the computation processing in step S**240**, the effect of the computation processing in step S**250**, the effect of the computation processing in step S**252**, and the effect of the computation processing based on the image sensor information **50**.
[0217] Steps S**250** and S**252** will now be described specifically. The predetermined subject image **30-3** is a field sequential image obtained by processing of combining a plurality of images captured by the monochrome image sensor **108** at a timing when light of each wavelength band is emitted in a case where light having a plurality of wavelength bands is sequentially emitted. For example, as illustrated in FIG. **45**, in step S**206-1** described above, processing including step S**250** generates a mosaic image. Processing including step S**252** generates a field sequential image again from the mosaic image, whereby the first training image **32-1** is generated. In step S**206-1** in FIG. **45**, processing other than steps S**250** and S**252** is not depicted.
[0218] FIG. **46** illustrates an example of the best focus simulation processing in the image data generation processing (step S**126**) illustrated in FIG. **43**. The best focus simulation processing illustrated in FIG. **43** and FIG. **46** can also be called step S**306**. FIG. **46** differs from FIG. **41** in that it further includes processing of generating a mosaic image from the predetermined subject image **30-3** (step S**350**) and processing of demosaicing the mosaic image (step S**352**). Further, step S**350** in FIG. **46** is processing similar to step S**250** in FIG. **44**, and step S**352** in FIG. **46** is processing similar to step S**252** in FIG. **44**. For example, the training device processing section **16** performs computation processing that appropriately combines the above step S**330-1**, the above step S**340**, step S**350**, step S**352**, and computation processing based on the image sensor information **50** not depicted in FIG. **46**. As a result, the training device processing section **16** can generate the true image **36**. Thus, step S**306** can be performed to obtain a computation processing result that reflects the effect of the computation processing in step S**330**, the effect of the computation processing in step S**340**, the effect of the computation processing in step S**350**, the effect of the computation processing in step S**352**, and the effect of the computation processing based on the image sensor information **50**. The true image **36** may be generated by processing in which steps S**330**, S**350**, and S**352** are omitted from the best focus simulation processing (step S**306**). In other words, the true image **36** may be generated by performing processing corresponding to step S**340** for the predetermined subject image **30-3**.
[0219] Based on the foregoing, in the information processing system **100** according to the present embodiment, the given imaging system **104** includes the monochrome image sensor **108**. The predetermined subject image **30-3** is a field sequential image obtained by processing of combining a plurality of images captured by the monochrome image sensor **108** at a timing when light of each wavelength band is emitted in a case where light having a plurality of wavelength bands is sequentially emitted. The first imaging system **101** includes the simultaneous-type image sensor **106** that has a plurality of pixels having colors different from each other and in which one color is allocated to each of the pixels. The defocus simulation processing (step S**206**) further includes processing of generating, from the predetermined subject image **30-3**, a mosaic image in which one color is allocated to each of the pixels, processing of demosaicing the mosaic image, processing of simulating the difference between the given imaging system **104** and the first imaging system **101**, and processing of reducing the predetermined subject image **30-3**. The processing of simulating the difference between the given imaging system **104** and the first imaging system **101** in the defocus simulation processing (step S**206**) is based on the transfer function or the point spread function at the object distance at which the given imaging system **104** is focused, and the transfer function or

the point spread function at a plurality of object distances of the first imaging system **101**. The true image **36** is an image generated by performing the best focus simulation processing (step S**306**) or an image generated by performing the processing of reducing the predetermined subject image **30-3**. The best focus simulation processing (step S**306**) further includes processing of generating a mosaic image, processing of demosaicing the mosaic image, processing of simulating the difference between the given imaging system **104** and the first imaging system **101**, and processing of reducing the predetermined subject image **30-3**. The processing of simulating the difference between the given imaging system **104** and the first imaging system **101** in the best focus simulation processing (step S**306**) is based on the transfer function or the point spread function at the object distance at which the given imaging system **104** is focused, and the transfer function or the point spread function at the object distance at which the first imaging system **101** is focused. In this way, in the case where the imaging method of the predetermined subject image **30** and the imaging method of the processing target image are different, a more appropriate data set including the training images **32** and the true image **36** can be generated.

[0220] Different trained models **120** may be used depending on imaging methods. In other words, in the information processing system **100** according to the present embodiment, for example, as illustrated in FIG. **47**, the memory section **110** may store a first trained model **121** and a second trained model **122**.

[0221] In a case where the memory section **110** stores the first trained model **121** and the second trained model **122**, the flow illustrated in FIG. **3** may be replaced, for example, by the flow in FIG. **48**. The processing section **130** reads a processing target image (step S**10**) and thereafter performs processing of checking an imaging method of the first imaging system **101** (step S**12**). If the imaging method is a field sequential method, reading the first trained model (step S**21**), correction processing (step S**31**), and corrected image output (step S**41**) are performed. On the other hand, if the imaging method is a Bayer simultaneous method, reading the second trained model (step S**22**), correction processing (step S**32**), and corrected image output (step S**42**) are performed. Steps S**21** and S**22** in FIG. **48** are processing corresponding to step S**20** in FIG. **3**. Similarly, steps S**31** and S**32** in FIG. **48** are processing corresponding to step S**30** in FIG. **3**, and steps S**41** and S**42** in FIG. **48** are processing corresponding to step S**40** in FIG. **3**.

[0222] In this case, step S**100** in FIG. **7** may be performed as in step S**101** in FIG. **49** and step S**102** in FIG. **50**. Specifically, first trained model creation processing (step S**101**) in FIG. **49** employs the image data generation processing in step S**124** in FIG. **39** for step S**100** in FIG. **7**. Similarly, second trained model creation processing (step S**102**) in FIG. **50** employs the image data generation processing in step S**126** in FIG. **43** for step S**100** in FIG. **7**.

[0223] Further, the technique of the present embodiment can also be applied to a case where the given imaging system **104** and the first imaging system **101** employ different observation methods. Referring to FIG. **51**, a technique of image data generation processing in the case where observation methods are different will be described. The image data generation processing in FIG. **51** can also be called step S**128**, and an original image for step S**128** can also be called a predetermined subject image **30-4**. Step S**128** in FIG. **51** differs from step S**124** in FIG. **39** in the contents of the defocus simulation processing (step S**208**) and the best focus simulation processing (step S**308**) and in that observation method information **60** is read before steps S**206** and S**306** are performed. The observation method information **60** is, for example, information regarding an observation method in the first imaging system **101**. In other words, in a case of the example in FIG. **51**, the observation method information **60** not depicted in FIG. **4** is further stored in the training device memory section **18**. In FIG. **51**, the second imaging system **102** is illustrated as a representative of the given imaging system **104**, which is the same as in the example in FIG. **39**.

[0224] The observation method can also be called an observation mode. The case where observation methods are different may be, for example, a case where light sources for observation are different, but may be, for example, a case where image processing techniques performed from when a user performs processing of capturing an image of a subject to when acquiring the predetermined subject image **30-4** are different. Examples of the observation method include a white light imaging (WLI) mode that uses white illumination light and a special light observation mode that uses special light which is not white light. The special light observation mode includes a narrow band imaging (NBI) mode that uses two types of narrow band light. The two types of narrow band light are narrow band light included in a blue wavelength band and narrow band light included in a green wavelength band. The image processing is different between the WLI and the NBI when a color image is generated from image signals output by the image sensor. For example, a content of the demosaicing processing or a parameter in the image processing is different. As the special light observation mode, for example, a red dichromatic imaging (RDI) mode may be employed. The RDI mode is an observation mode using narrow band light included in an umber wavelength band, narrow band light included in a green wavelength band, and narrow band light included in a red wavelength band. For example, the technique disclosed in U.S. Pat. No. 9,775,497 B**2** is used.

[0225] FIG. **52** illustrates an example of defocus simulation processing (step S**208-1**) of generating the first training image **32-1** from the predetermined subject image **30-4**. Step S**208-1** in FIG. **52** differs from step S**204-1** in FIG. **50** in that it further includes WLI mode processing (step S**262**), NBI mode processing (step S**264**), RDI mode processing (step S**266**), and TXI mode processing (step S**268**). TXI is an abbreviation of texture and color enhancement imaging, which will be detailed later.

[0226] Step S**128** in FIG. **51** is an example in which processing and the like that is the different point described above is added to S**124** in FIG. **39**, but not limited thereto. For example, processing and the like that is the different point described above may be added to step S**126** in FIG. **43**. In this case, although not depicted in the drawings, the color shift determination processing (step S**190**) in FIG. **43** is further performed before steps S**208** and S**308** are performed. Step S**208** in FIG. **52** in this case further includes steps S**240**, S**250**, and S**252** in FIG. **44**. Similarly, step S**308** in FIG. **53** in this case further includes steps S**340**, S**350**, and S**352** in FIG. **46**. In the following, a description that overlaps with the description of step S**124** in FIG. **39** and step S**126** in FIG. **43** is omitted if appropriate.

[0227] For example, although not illustrated in a flowchart, the training device processing section **16** reads the observation method information **60** and acquires an observation method used for the first imaging system **101**. The training device processing section **16** then selects any of steps S**262**, S**264**, S**266**, and S**268** as processing corresponding to the acquired observation method.

[0228] For example, in a case where the first imaging system **101** captures an image in the TXI mode, information indicating this is stored as observation method information **60** in the training device memory section **18**. The training device processing section **16** then reads the observation method information **60** to perform the defocus simulation processing (step S**208**) including the TXI mode processing (step S**368**) for the predetermined subject image **30-4**. Specifically, for example, the training device processing section **16** performs processing of decomposing the predetermined subject image **30-4** into a texture image portion which is an image portion associated with a surface structure of the predetermined subject image **30-4**, and a base image portion other than the texture image portion. The training device processing section **16** then performs first processing of enhancing the surface structure associated with the texture image portion, second processing of optimizing brightness of the base image portion, and third processing of optimizing a color tone of an image that combines an image associated with the first processing and an image associated with the second processing. This can result in the training image **32** that simulates the effect of imaging in the TXI mode for the predetermined subject image **30-4**. As a result, machine learning can be performed with a data set including more accurate training images **32**.

[0229] Further, for example, although not depicted in the drawings, in a case where the first imaging system **101** captures an image in the WLI mode or the NBI mode, information indicating this is stored as observation method information **60** in the training device memory section **18**. The training device processing section **16** then reads the observation method information **60** to perform color complementation to match a light source for the predetermined subject image **30-4**. The color complementation may be performed, for example, in addition to step S**252** in FIG. **44**. For example, when the WLI mode processing (step S**262**) is selected, the training device processing section **16** performs processing of complementing an R image and a B image using a G image, in addition to step S**252**. For example, when the NBI mode processing (step S**264**) is selected, the training device processing section **16** performs processing of complementing a G image and a B image independently of each other, in addition to step S**252**.

[0230] FIG. **53** illustrates an example of the best focus simulation processing (step S**308**) of generating the true image **36** from the predetermined subject image **30-4** in the image data generation processing (step S**128**). Step S**308** in FIG. **53** differs from step S**304** in FIG. **41** in that it further includes WLI mode processing (step S**362**), NBI mode processing (step S**364**), RDI mode processing (step S**366**), and TXI mode processing (step S**368**). Step S**362** in FIG. **53** is processing similar to step S**262** in FIG. **52**, step S**364** in FIG. **53** is processing similar to step S**264** in FIG. **52**, step S**366** in FIG. **53** is processing similar to step S**266** in FIG. **52**, and step S**368** in FIG. **53** is processing similar to step S**268** in FIG. **52**. The true image **36** may be generated by processing in which step S**330** or the like is omitted from the best focus simulation processing (step S**308**) in FIG. **53**.

[0231] Further, the correction table **1400** in FIG. **25** can be created, for example, based on the result of evaluation of frequency characteristic for each object distance for contrast or amplitude of various image data. For example, in FIG. **54**, input data indicated by K**1** is input to the trained model **1120**, and output data indicated by K**11** is output from the trained model **1120**. Similarly, input data indicated by K**9** is input to the trained model **1120**, and output data indicated by K**19** is output from the trained model **1120**.

[0232] Although the image data associated with the input data is an image of a chart denoted by K**2** in FIG. **54**, it may be an image of biological tissue or an image of a subject that imitates biological tissue as described above, or any image that is equivalent to the training image **32**. The contrast of the image of the chart may be common to the contrast of biological tissue as described above. The input data in FIG. **54** may be an actual image of the chart or an image created by simulation. Further, K**3** in FIG. **54** is a conceptual representation of the spread function corresponding to the chart indicated by K**2**. In output data indicated by K**11**, the chart image indicated by K**12** corresponds to the output image **134**, and K**13** is a conceptual representation of the spread function corresponding to the chart indicated by K**12**.

[0233] In FIG. **54**, the input data on the upper side of the drawing sheet indicates that it includes a chart image with lower frequencies, and the input data on the lower side of the drawing sheet indicates that it includes a chart image with higher frequencies. For example, the frequency of the image of the chart indicated by K**2** associated with the input data of K**1** is lower than the frequency of the image of the chart associated with the input data of K**9**. Due to the effect of the correction processing (step S**1030**), the image of the chart associated with the output data of K**11** is sharper than the image of the chart indicated by K**2**. Therefore, the user can quantitatively evaluate the difference in contrast between the training image **32** and the output image **134**, or the difference in amplitude between the training image **32** and the output image **134**, by analyzing the spread function indicated by K**13** and the spread function indicated by K**3**.

[0234] On the other hand, for example, when the input data indicated by K**9** in FIG. **54** is compared with the output data indicated by K**19**, the change in contrast or the change in amplitude by the correction processing (step S**1030**) is extremely small. The frequency region associated with the input data indicated by K**9** in FIG. **54** is, for example, the frequency region indicated by B**1** in FIG. **27**, where the value of MTF decreases to 0. In this way, at higher frequencies, it is difficult to accurately estimate a significant difference by the effect of the correction processing (step S**1030**).

Therefore, in such a frequency range, blur correction need not be performed in the blur adjustment processing (step S**1032**).
[0235] The frequency characteristic evaluation illustrated in FIG. **54** is performed for each object distance to produce, for example, a frequency characteristic diagram indicated by K**20** or K**30** in FIG. **55**. The vertical axis is not specified, but may be contrast or amplitude. The graph of K**20** illustrates a frequency characteristic when the object distance is the best focus distance. In the graph of K**20**, K**21** is a first frequency characteristic, which is the frequency characteristic of the output image **134** obtained by the correction processing (step S**1030**). Further, K**23** is a frequency characteristic of the training image **32** before the correction processing (step S**1030**) is performed.
[0236] K**22** in the graph of K**20** in FIG. **55** is a target frequency characteristic by the effect of the blur adjustment processing (step S**1032**), which is the frequency characteristic of the true image **36**. More specifically, K**22** is a plot of the result of frequency characteristic evaluation performed for an image generated by performing the best focus simulation processing (step S**300**) for the predetermined subject image **30** described above with reference to FIG. **10**. Alternatively, K**22** may be a plot of the result of frequency characteristic evaluation performed for the predetermined subject image **30** itself which is captured by the given imaging system **104** in focus as described above with reference to FIG. **11**.
[0237] In the frequency characteristic diagram K**20**, since an image is captured at an object distance that is the best focus distance, the training image **32** is substantially the same as the true image **36**, and the plots K**21**, K**22**, and K**23** overlap each other. For convenience, in the graph K**20**, the plots K**21**, K**22**, and K**23** are depicted slightly offset from each other.
[0238] In the frequency characteristic diagram K**30**, K**31** is a first frequency characteristic which is the frequency characteristic of the output image **134** when an image is captured at an object distance other than the best focus distance, and K**33** is a frequency characteristic of the training image **32**. On the other hand, K**32** is plotted by converting data of a second frequency characteristic indicated by K**22** into the frequency characteristic diagram K**30**.
[0239] Then, for example, a value of the correction factor at a frequency indicated by K**40** is determined, based on a value on the horizontal axis indicated by K**41**, a value on the horizontal axis indicated by K**42**, and the like, whereby the correction factor at a desired object distance and frequency, that is, one record in the correction table **1400** in FIG. **25**, can be obtained. Therefore, the correction table **1400** can be created by repeating the same technique as many as the number of combinations of the number of object distance data and the number of frequency data as necessary.
[0240] Based on the foregoing, in the information processing system **1000** according to the present embodiment, the correction table **1400** is a table in which the filter characteristic in the blur adjustment processing for matching the first frequency characteristic which is the frequency characteristic of the output image **134** to the second frequency characteristic which is a target frequency characteristic in the blur adjustment processing is associated with each of a plurality of object distances. The frequency characteristic is a function that quantitatively represents the relationship of contrast or amplitude to frequency.
[0241] In this way, the blur adjustment processing (step S**1032**) can be performed using the correction table **1400** including appropriate correction factors. Specifically, the blur adjustment processing (step S**1032**) is performed for the output image **134** based on the correction table **1400** created in this way, and the frequency characteristic based on the resulting blur adjustment processed image **1034** is as indicated by K**32**. As described above, since the frequency characteristic indicated by K**32** is a frequency characteristic based on the true image **36**, the output image **134** can be corrected to an image equivalent to the true image **36** by further performing the blur adjustment processing (step S**1032**) for the output image **134**.
[0242] Further, in the information processing system **1000** according to the present embodiment, the second frequency characteristic may be created by converting information of an image obtained by capturing an image of a subject in focus by the given imaging system, or an image generated by

the best focus simulation processing, into a frequency characteristic at an object distance other than the object distance at which the given imaging system is focused. In this way, a target frequency characteristic can be set as appropriate at an object distance other than the object distance that achieves focus. As a result, values of the correction factors in the correction table **1400** can be set as appropriate.

[0243] Further, in the information processing system **1000** according to the present embodiment, the training image **32** may be an image of biological tissue or an image of a subject that imitates biological tissue. In this way, the frequency characteristic can be evaluated using a subject with a contrast close to that of an actual subject to be imaged. As a result, blur correction can be performed appropriately for a processing target image with a contrast close to that of an actual subject to be imaged.

[0244] Further, in the correction table **1400** in the information processing system **1000** according to the present embodiment, the filter characteristic for blur correction associated with an object distance at which the MTF is zero at a frequency equal to or lower than the Nyquist frequency may be a filter characteristic that does not correct a blur for a frequency equal to or higher than a frequency at which the MTF is zero. In this way, appropriate blur correction processing can be performed for an image having frequency components in which a blur correction effect is obvious, among processing target images.

[0245] In the above example, the correction processing (step S**1030**) is performed for the whole of one processing target image. However, the technique of the present embodiment is not limited to this. For example, the processing target image may be divided and the correction processing (step S**1030**) may be performed for each of the divided processing target images.

[0246] Specifically, for example, it is assumed that a processing target image indicated by J**0** in FIG. **56** is input to the input section **1140** of the information processing system **1000**. In the correction processing (step S**1030**), the processing section **1130** performs processing of dividing the processing target image indicated by J**0** into four images, namely, an image indicated by J**11**, an image indicated by J**12**, an image indicated by J**13**, and an image indicated by J**14**. The image indicated by J**11** corresponds to a region indicated by J**1** in the original processing target image. The image indicated by J**12** corresponds to a region indicated by J**2** in the original processing target image. The image indicated by J**13** corresponds to a region indicated by J**3** in the original processing target image. The image indicated by J**14** corresponds to a region indicated by J**4** in the original processing target image. Although FIG. **56** illustrates an example in which a processing target image is divided into four images, the number of divided images is not limited to four, but can be determined by a user as appropriate.

[0247] The processing section **1130** then performs correction processing (step S**1030**) for the images indicated by J**11**, J**12**, J**13**, and J**14** in sequence. In other words, the processing section **1130** reads the trained model **1120** from the memory section **1110** and inputs the image indicated by J**11** to the trained model **1120**. As a result, an image indicated by J**21** and the estimated object distance label **1070** indicating an estimated object distance ED**1** are output from the trained model **1120**. Similarly, the processing section **1130** reads the trained model **1120** from the memory section **1110** and inputs the image indicated by J**12** to the trained model **1120**. As a result, an image indicated by J**22** and the estimated object distance label **1070** indicating an estimated object distance ED**2** are output from the trained model **1120**. Similarly, the processing section **1130** reads the trained model **1120** from the memory section **1110** and inputs the image indicated by J**13** to the trained model **1120**. As a result, an image indicated by J**23** and the estimated object distance label **1070** indicating an estimated object distance ED**3** are output from the trained model **1120**. Similarly, the processing section **1130** reads the trained model **1120** from the memory section **1110** and inputs the image indicated by J**14** to the trained model **1120**. As a result, an image indicated by J**24** and the estimated object distance label **1070** indicating an estimated object distance ED**4** are output from the trained model **1120**.

[0248] The images indicated by J**21**, J**22**, J**23**, and J**24** can be regarded as divided images of the output image **134** obtained by performing the correction processing (step S**1030**) for the subject image indicated by J**0**. The processing section **1130** then performs processing of combining the images indicated J**21**, J**22**, J**23**, and J**24** into the output image **134** indicated by J**30**. A region indicated by J**31** in the output image **134** indicated by J**30** corresponds to the image indicated by J**21**. Similarly, a region indicated by J**32** in the output image **134** indicated by J**30** corresponds to the image indicated by J**22**, a region indicated by J**33** corresponds to the image indicated by J**23**, and a region indicated by J**34** corresponds to the image indicated by J**24**.
[0249] Further, for example, information on the estimated object distance ED**1** corresponding to the image J**21** may be superimposed and displayed on the region indicated by J**31**. Similarly, information on the estimated object distance ED**2** corresponding to the image J**22** may be superimposed and displayed on the region indicated by J**32**, information on the estimated object distance ED**3** corresponding to the image J**23** may be superimposed and displayed on the region indicated by J**33**, and information on the estimated object distance ED**4** corresponding to the image J**24** may be superimposed and displayed on the region indicated by J**34**.
[0250] In addition to the correction processing (step S**1030**), the blur adjustment processing (step S**1032**) may be further performed for each of the divided processing target images. In FIG. **57**, the processing section **1130** performs processing of dividing the processing target image indicated by J**0** in the correction processing (step S**1030**) into the images indicated by J**11**, J**12**, J**13**, and J**14**, in the same manner as in FIG. **56**. The processing section **1130** then performs the correction processing (step S**1030**) for each of the images indicated by J**11**, J**12**, J**13**, and J**14**.
[0251] Based on the foregoing, in the information processing system **100** according to the present embodiment, the processing section **1130** estimates an object distance for each predetermined divided region of a processing target image, acquires from the correction table **1400** a filter characteristic associated with the object distance estimated for each divided region, and performs blur adjustment processing for the output image **134** using the acquired filter characteristic. In this way, the blur adjustment processing can be performed according to the object distance estimated for each divided region. As a result, the blur adjustment processing can be performed in consideration of the flatness of a subject.
[0252] The processing in step S**1034** in FIG. **24** will be described more specifically. FIG. **57** is a conceptual illustration of the processing in step S**1034**. In FIG. **57**, K**50** is a conceptually simplified illustration of the filter characteristic as a real-space filter, which is inverse Fourier transform of the frequency response function indicated by H**12** in FIG. **25**. In other words, step S**1034** includes processing of performing inverse Fourier transform of the frequency response function and processing of performing convolution of a real-space filter obtained by the inverse Fourier transform for the output image **134**. Since the image obtained by convolution corresponds to the blur adjustment processed image **1034**, the processing section **1130** outputs the blur adjustment processed image **1034** as a corrected image from the output section **1150** in step S**1040** in FIG. **18**. In this way, in the information processing system **1000** according to the present embodiment, the filter characteristic is a real-space filter, and the processing section **1130** performs blur adjustment processing by performing convolution of the real-space filter for the output image **134**. In this way, the blur adjustment processing in real space can be performed.
[0253] Further, the processing section **1130** may perform step S**1034** in FIG. **24** in frequency space. FIG. **58** is a conceptual illustration of the processing in step S**1034** performed in frequency space. In FIG. **58**, K**60** is a conceptually simplified illustration of a frequency signal obtained by Fourier transform of the output image **134**. Further, K**61** in FIG. **58** is a conceptually simplified illustration of a frequency signal obtained by multiplying the frequency signal of K**60** by the frequency response function indicated by H**12** in FIG. **25**. In other words, step S**1034** includes processing of performing Fourier transform of the output image **134**, processing of multiplying a frequency signal obtained by the Fourier transform by the frequency response function as the filter

characteristic obtained in step S**1033**, and processing of performing inverse Fourier transform of a frequency signal obtained by the multiplication. Since the image obtained by inverse Fourier transform corresponds to the blur adjustment processed image **1034**, the processing section **1130** outputs the blur adjustment processed image **1034** as a corrected image from the output section **1150** in step S**1040** in FIG. **18**.

[0254] Based on the foregoing, in the information processing system **1000** according to the present embodiment, the filter characteristic is a frequency response function, and the processing section **1130** performs blur adjustment processing by performing Fourier transform of the output image **134**, multiplying a frequency signal acquired by the Fourier transform by the frequency response function, and performing inverse Fourier transform of a frequency signal acquired by the multiplication. In this way, the blur adjustment processing in frequency space can be performed.

[0255] The technique in FIG. **57** and the technique in FIG. **58** are mathematically equivalent. Therefore, the blur adjustment processed **1034** obtained by the technique in FIG. **57** is substantially the same as the blur adjustment processed image **1034** obtained by the technique in FIG. **58**. Whether to perform step S**1034** by the technique in FIG. **57** or to perform step S**1034** by the technique in FIG. **58** can be determined as appropriate by a user.

[0256] For example, the trained model **1120** according to the present embodiment may be constructed with two or more neural networks. Specifically, for example, as illustrated in FIG. **59**, the trained model **1120** may include a first input layer, a second input layer, a first intermediate layer, a second intermediate layer, a first output layer, and a second output layer. For example, a processing target image is input to the first input layer. The first intermediate layer performs computation for an output from the first input layer. The first output layer then generates the output image **134** from an output from the first intermediate layer. Further, the processing target image is input to the second input layer. The second intermediate layer performs computation for an output from the second input layer. The second output layer then generates the estimated object distance label **1070** from an output from the second intermediate layer. Whether to construct the trained model **1120** as illustrated in FIG. **23** or as illustrated in FIG. **59** can be determined as appropriate by a user.

[0257] Although the embodiments have been described in detail above, it will be readily understood by those skilled in the art that many modifications can be made without substantially departing from the novel matters and effects in the present disclosure. Therefore, all such modifications are intended to be included within the scope of the present disclosure. Any term cited with a different term having a broader meaning or the same meaning at least once in the specification and the drawings can be replaced by the different term in any place in the specification and the drawings. All combinations of the embodiments and modifications are also included in the scope of the present disclosure. The configuration, operation, and the like of the information processing system, the endoscope system, the image processing method, and the program are not limited to those described in the embodiments and can be modified in various ways.

Claims

1. An information processing system comprising: a memory configured to store a trained model trained by machine learning with a data set including a training image, a true image, and an object distance label, and a correction table in which an object distance is associated with a filter characteristic for blur correction, the object distance being a distance between an imaging system and a subject; and one or more processors, wherein the training image is generated by performing defocus simulation processing that simulates, for a predetermined subject image in focus captured by a given imaging system, an effect of a blur caused by defocus of a first imaging system, based on a transfer function or a point spread function of the first imaging system at a predetermined object distance, the defocus simulation processing is performed for a region on an optical axis of the first imaging system and a region other than on the optical axis in the training image, based on the transfer function or the point spread function on the optical axis, the true image is an image generated by performing best focus simulation processing that simulates, for the predetermined subject image, a state in which the first imaging system is focused, based on the transfer function or the point spread function at the object distance at which the first imaging system is focused, or the predetermined subject image itself, the trained model is trained by machine learning so that the training image is the true image, and trained by machine learning by applying, as the object distance label, the object distance of the transfer function or the point spread function of the first imaging system used in the defocus simulation processing, and the one or more processors use the trained model to generate an output image in which a blur of a processing target image is corrected, the processing target image being an image captured by the first imaging system, estimate the object distance of the processing target image, and acquire, from the correction table, the filter characteristic associated with the estimated object distance, and perform blur adjustment processing for the output image using the acquired filter characteristic.

2. The information processing system according to claim 1, wherein the correction table is a table in which the filter characteristic in the blur adjustment processing for matching a first frequency characteristic to a second frequency characteristic is associated with each of a plurality of the object distances, the first frequency characteristic being a frequency characteristic of the output image, the second frequency characteristic being a target frequency characteristic in the blur adjustment processing, and the frequency characteristic is a function that quantitatively represents a relationship of contrast or amplitude to frequency.

3. The information processing system according to claim 2, wherein the second frequency characteristic is created by converting information of an image obtained by capturing an image of a subject in focus by the given imaging system, or an image generated by the best focus simulation processing, into the frequency characteristic at the object distance other than the object distance at which the given imaging system is focused.

4. The information processing system according to claim 2, wherein the filter characteristic is a frequency response function, and the one or more processors perform the blur adjustment processing by performing Fourier transform for the output image, multiplying a frequency signal acquired by the Fourier transform by the frequency response function, and performing inverse Fourier transform of a frequency signal acquired by the multiplication.

5. The information processing system according to claim 2, wherein the filter characteristic is a real-space filter, and the one or more processors perform the blur adjustment processing by performing convolution of the real-space filter for the output image.

6. The information processing system according to claim 1, wherein the trained model is constructed with one neural network, the neural network includes an input layer to which the processing target image is input, an intermediate layer configured to perform computation for an output from the input layer, a first output layer configured to generate the output image from an output from the intermediate layer, and a second output layer configured to estimate the object distance from an output from the intermediate layer.

7. The information processing system according to claim 1, wherein the one or more processors estimate the object distance for each of predetermined divided regions of the processing target image, and acquire, from the correction table, the filter characteristic associated with the estimated object distance for each of the divided regions, and perform the blur adjustment processing for the output image using the acquired filter characteristic.

8. The information processing system according to claim 1, wherein the training image is an image of biological tissue, or an image of a subject that imitates the biological tissue.

9. The information processing system according to claim 1, wherein in the correction table, the filter characteristic for the blur correction associated with the object distance at which a modulation transfer function is zero at a frequency equal to or lower than Nyquist frequency is the filter characteristic that does not correct the blur for a frequency equal to or higher than a frequency at which the modulation transfer function is zero.

10. An endoscope system comprising: the information processing system according to claim 1; and an endoscopic scope configured to capture the processing target image.

11. An image processing method using a trained model trained by machine learning with a data set including a training image, a true image, and an object distance label, and a correction table in which an object distance is associated with a filter characteristic for blur correction, the object distance being a distance between an imaging system and a subject, wherein the training image is generated by performing defocus simulation processing that simulates, for a predetermined subject image in focus captured by a given imaging system, an effect of a blur caused by defocus of a first imaging system, based on a transfer function or a point spread function of the first imaging system at a predetermined object distance, the defocus simulation processing is performed for a region on an optical axis of the first imaging system and a region other than on the optical axis in the training image, based on the transfer function or the point spread function on the optical axis, the true image is an image generated by performing best focus simulation processing that simulates, for the predetermined subject image, a state in which the first imaging system is focused, based on the transfer function or the point spread function at the object distance at which the first imaging system is focused, or the predetermined subject image itself, the trained model is trained by machine learning so that the training image is the true image, and trained by machine learning by applying, as the object distance label, the object distance of the transfer function or the point spread function of the first imaging system used in the defocus simulation processing, the image processing method comprising: using the trained model to generate an output image in which a blur of a processing target image is corrected, the processing target image being an image captured by the first imaging system; estimating the object distance of the processing target image; and acquiring, from the correction table, the filter characteristic associated with the estimated object distance, and performing blur adjustment processing for the output image using the acquired filter characteristic.

12. A non-transitory information storage medium that stores a program for causing a computer to execute the image processing method according to claim 11.